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Review

The brittle-ductile transition in porous rock: A review

Teng-fong Wong [a], Patrick Baud [b,*]

[a] Department of Geosciences, State University of New York at Stony Brook, NY 11794-2100, USA
[b] Institut de Physique du Globe de Strasbourg (UMR 7516 CNRS, Université de Strasbourg/EOST), Strasbourg, France

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ABSTRACT

Many of the earliest laboratory studies of the brittle-ductile transition were on porous rocks, with a focus on the evolution of failure mode from brittle faulting to cataclastic flow with increasing pressure. Recent advances in this area are reviewed. Porosity has been demonstrated to exert critical control on the brittle-ductile transition, and its phenomenology has two common attributes. Under low confinement, brittle faulting develops as a dilatant failure mode. Under high confinement, delocalized cataclasis is accompanied by shear-enhanced compaction and strain hardening. Plasticity models such as the cap and critical state models have been developed to describe such constitutive behaviors, and many aspects of the laboratory data on porous rock have been shown to be in basic agreement. Bifurcation analysis can be used in conjunction with a constitutive model to predict the onset of strain localization, which is in qualitative agreement with the laboratory data. However, recent studies have also underscored certain complexities in the inelastic behavior and failure mode. In some porous sandstones, compaction bands would develop as a localized failure mode intermediate between the end members of brittle faulting and cataclastic flow. In limestones (and selected sandstones) under relatively high confinement, cataclastic flow is accompanied first by shear-enhanced compaction which then evolves to dilatancy. Various techniques have been employed to characterize the microstructure and damage, which have elucidated the deformation mechanisms associated with the brittle-ductile transition. These observations have revealed a diversity of micromechanical processes, and fundamental differences were observed especially between sandstone and limestone with regard to inelastic compaction. Micromechanical models that have been formulated to describe these processes include the pore-emanated and sliding wing crack models in the brittle faulting regime, and the Hertzian fracture and cataclastic pore collapse models in the cataclastic flow regime. Numerical techniques based on the discrete element method have also been employed to simulate these processes. Comparison of the model predictions with laboratory and microstructural observations has provided useful insights into the mechanics of brittle-ductile transition in porous rock.

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1. Introduction

In experimental rock deformation, the inelastic behavior is commonly categorized into the brittle and ductile fields. The distinction between brittle and ductile behavior is in the first place a macroscopic one, depending on whether or not the rock is capable of undergoing a substantial permanent strain without macroscopic fracture (Paterson and Wong, 2005). Because such a categorization is essentially macroscopic, it does not hinge on the specification of operative deformation mechanisms (Rutter, 1986; Evans et al., 1990). The distinction between these two fields is not in all respects a sharp one, especially when microscopic processes are also taken into account, but it is nevertheless a very useful one.

The deformation mechanism associated with macroscopic fracture in the brittle field is likely to be predominately cataclastic, involving microcracking and frictional sliding among the grain fragments. In contrast, inelastic deformation in the ductile regime may arise from the complex interplay of multiple deformation mechanisms, including crystal plasticity, diffusional mass transfer and cataclasis (Evans et al., 1990; Blenkinsop, 2000; Passchier and Trouw, 2005; Paterson and Wong, 2005). The sensitivity of each of these deformation mechanisms to pressure, temperature, strain rate and presence of fluid can be quite different. Ductile flow commonly develops in a delocalized manner, but strain localization may also develop due to mechanisms including strain, thermal, geometric and structural softening associated with a rheological

* Corresponding author. Fax: +33 368850083.
E-mail address: patrick.baud@unistra.fr (P. Baud).

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response that is sensitive to strain rate and grain size (Poirier, 1980; Evans and Wong, 1985; Paterson and Wong, 2005).

The failure mode of a porous rock undergoes a transition from brittle faulting to ductile flow with increases in pressure and temperature. In earlier laboratory studies of the brittle-ductile transition (von Karman, 1911; Paterson, 1958; Heard, 1960; Handin et al., 1963; Mogi, 1966; Byerlee, 1968), the operative deformation mechanisms were primarily cataclastic, such that the failure mode would switch from brittle faulting (localized cataclasis) to ductile flow by distributed cataclasis as confining pressure was increased at constant temperature. The latter is also referred to as "cataclastic flow". It is now appreciated that this transition from brittle faulting to calaclastic flow is intimately related to the initial porosity and how it evolves with stress. The porosity evolution can be dilatant or compactant. Whereas dilatancy is commonly observed as a precursor to brittle faulting in both compact and porous rocks (Brace, 1978; Paterson and Wong, 2005), failure by cataclastic flow may be accompanied by a positive or negative change in volume depending on the initial porosity of the rock.

In a compact rock such as Carrara marble (with an initial porosity of 0.7%), Fredrich et al. (1989) observed significant volume increase in their samples undergoing cataclastic flow. This dilatant behavior is fundamentally different from that in a porous rock, which typically undergoes inelastic compaction during cataclastic flow (Edmond and Paterson, 1972; Brace, 1978; Jamison and Teufel, 1979; Carroll, 1980; Wong et al., 1997). It is difficult to pinpoint precisely where this demarcation in porosity evolution behavior between a compact and porous rock lies. In one of the earliest reviews on this issue, Brace (1978) suggested a nominal cutoff porosity of 5%, above which the rock would be considered to be "porous", in the sense that cataclastic flow would be compactant. However, a later study by Baud et al. (2000a) documented the development of compactant cataclastic flow in Solnhofen limestone with an initial porosity as low as 3%.

In this review we will focus on the transition from brittle faulting to compactant cataclastic flow in porous rock. Given these earlier observations on the brittle-ductile transition in a porous rock, for a long time the implicit assumption was that, whereas strain localization (in the form of shear bands) would readily develop in the brittle field, it is inhibited in a porous rock undergoing inelastic compaction which would presumably accumulate strain only in a delocalized manner. However, in the past two decades a series of field and laboratory observations in porous sandstone have demonstrated that this conception of the brittle-ductile transition is overly simplistic, in that we now recognize that there can be an intermediate regime with a failure mode associated with the localized development of compaction bands. These bands are planar structures that have undergone cataclastic compaction due to shortening in a direction subperpendicular to their planar surfaces with negligible shear. Compaction bands were first documented in the field (Hill, 1989; Mollema and Antonellini, 1996), and then explored systematically in theoretical, laboratory and field studies (e.g., Olsson, 1999; Olsson and Holcomb, 2000; Issen and Rudnicki, 2000; Wong et al., 2001; Baud et al., 2004; Sternlof et al., 2005; Fortin et al., 2006; Eichhubl et al., 2010; Schultz et al., 2010; Fossen et al., 2011; Baud et al., 2012).

The brittle-ductile transition in porous rock is a topic of importance in many geological applications, including the tectonics of faulting and deformation band formation (Aydin and Johnson, 1978; Jamison and Stearns, 1982; Underhill and Woodcock, 1987; Shipton and Cowie, 2001), the coupling between strain localization and fluid flow (Antonellini and Aydin, 1994; Fowles and Burley, 1994; Knipe, 1997; Wong and Zhu, 1999), tectonic evolution and fluid flow in sedimentary formation (Casey and Butler, 2004; Bjorlykke et al., 2005; Sheldon et al., 2006), reservoir compaction and subsidence (Boutéca et al., 1996; Fisher et al., 1999; Nagel, 2001; Makowitz and Milliken, 2003), borehole instability and well failure (Veeken et al., 1989; Fredrich et al., 2000; Coelho et al., 2005), seismic attenuation (Yarushina and Podladchikov, 2010), and physical limit on ground motion associated with earthquake rupture (Andrews et al., 2007; Andrews, 2007). Our review will be from the perspective of the rock deformation laboratory, but attempts will be made to establish connection with the field observations, so as to gain complementary insights and a deeper understanding of the mechanics of brittle-ductile transition.

Since in porous rock this transition of failure mode involves primarily cataclastic deformation mechanisms, which are sensitively dependent on pressure but relatively insensitive to temperature, it is also described as the "low-temperature brittle-ductile transition" (Rutter and Hadizadeh, 1991). It should be noted that under elevated temperatures and pressures, thermally activated deformation mechanisms related to crystal plasticity and diffusive mass transfer will become predominant (with the role of cataclasis reduced to being secondary), leading ultimately to homogeneous plastic flow. In this scenario there are two sequential stages in the brittle-ductile transition, with cataclastic flow as the intermediate step in the transition from brittle faulting to plastic flow (Tullis and Yund, 1992; Hirth and Tullis, 1994; Paterson and Wong, 2005; Passchier and Trouw, 2005). Although the brittle-plastic transition is also of importance in many geological and geophysical problems (Kohlstedt et al., 1995; Scholz, 2002) it is beyond the scope of the present review.

The organization of this review is as follows. First we will review critically the mechanical data and macroscopic observations on failure mode. The more systematic studies have been on sedimentary rocks including sandstone and limestone, but we will also discuss data on other porous rocks such as chalk and tuff. Common attributes of the phenomenology of the brittle-ductile transition in various porous rocks will be highlighted, and some of the fundamental differences with the behavior in a compact crystalline rock will be emphasized. We will then review the constitutive models that have been proposed to capture the deformation and failure behavior. In particular, we will focus on two popular plasticity models: the critical state model (Schofield and Wroth, 1968; Muir Wood, 1990) and cap model (DiMaggio and Sandler, 1971; Desai and Siriwardane, 1984; Lubarda et al., 1996).

Even though the phenomenological behaviors in porous rock and soil are qualitatively similar, an uncritical application of concepts such as critical state soil mechanics to porous rock may lead to overly simplistic interpretations, especially with regard to the operative deformation mechanism and damage mechanics. We will next review microstructural observations on the deformed and failed samples, which have provided useful insights into the deformation mechanisms and micromechanics of damage accumulation and failure. A key conclusion drawn from these observations is that even though the phenomenological behaviors of most porous rocks are qualitatively similar, a diversity of micromechanical processes are actually responsible for the macroscopic mechanical responses. Microstructural observations have documented that the cataclastic mechanisms operative in a clastic rock are significantly different from those in a carbonate rock. Unlike critical state soil mechanics in which porosity is the dominant parameter, additional microstructural parameters including the grain and pore sizes have proved to be as important in controlling the micromechanics of grain crushing and pore collapse in rock.

Motivated by the microstructural observations, a number of micromechanical models have been proposed to capture the brittle and ductile failure in porous rock. In the brittle field, we will discuss the pore-emanated crack (Sammis and Ashby, 1986) and sliding wing crack (Horii and Nemat-Nasser, 1986; Ashby and Sammis,

1990; Kemeny and Cook, 1991) models. In relation to these models, analytic estimates of the brittle strength as a function of the initial damage have recently been derived (Zhu, 2010; Zhu et al., 2011), and the theoretical predictions can conveniently be compared with laboratory data. In the ductile field, we will discuss the pore collapse (Curran and Carroll, 1979; Zhu et al., 2010) and Hertzian fracture (Zhang et al., 1990a; Wong et al., 1997) models, which have been developed for analyses of grain crushing and pore collapse in carbonate and siliciclastic rocks, respectively.

A key question on the mechanics of the brittle-ductile transition is: what are the intrinsic microstructural attributes and extrinsic loading conditions critical for the onset and development of strain localization? Important insights into the mechanics of strain localization have been gained from the continuum analysis of its inception as a bifurcation in the constitutive response of a porous medium (Rudnicki and Rice, 1975; Issen and Rudnicki, 2000; Bésuelle and Rudnicki, 2004). There have also been attempts to use the network or discrete element methods to tackle this issue (Katsman et al., 2005; Wang et al., 2008; Marketos and Bolton, 2009). While the laboratory data can provide useful physical insights into how strain localization would develop under controlled conditions, they are limited by the sample scale and number of deformation bands that can develop. Therefore an important challenge has been the scaling and extrapolation of laboratory data to the field. We will discuss the related issues, especially the application of fracture mechanics to analyze the propagation of deformation bands.

2. Phenomenology of the brittle-ductile transition in porous rock

One of the first systematic studies of the control of volume change on failure mode was by Edmond and Paterson (1972), who conducted triaxial compression experiments on porous materials including sandstone, limestone, talc and graphite, and contrasted their behavior with compact materials such as sodium chloride and Carrara marble. In an insightful review of such data, Brace (1978) pointed to the necessity of adopting fundamentally different frameworks for compact and porous rocks in analyzing their inelastic behaviors and failure modes. In relation to porous rock, Brace (1978) emphasized the usefulness of systematically acquiring mechanical data for not only nonhydrostatic loading, but also hydrostatic loading. He elaborated on the mechanical interpretation of the sigmoidal-shaped hydrostat of a porous rock, with an inflection point that marks the critical pressure for the onset of pore collapse. Brace (1978) also highlighted an effective methodology that had been adopted at Lawrence Livermore Laboratory and Terra Tek, whereby critical information on the onset and development of inelastic compaction can be derived by comparing the hydrostatic and nonhydrostatic data (Schock et al., 1973; Bhatt et al., 1975; Johnson and Green, 1976; Schatz, 1976). This remains the methodology and mechanical framework that have provided the experimentalist with some of the most consistent and useful data on inelastic compaction of porous rock and the brittle-ductile transition.

2.1. Mechanical deformation

We will adopt the convention that compressive stresses and compactive strains (i.e. shortening and porosity decrease) are positive. Most of the mechanical data reviewed here were acquired under conventional triaxial compression, with the maximum and minimum principal stresses denoted by σ_1 and σ_3, respectively. For a saturated sample deformed under drained condition, the pore pressure will be denoted by p_p, and the difference between confining and pore pressures will be referred to as the "effective pressure". Likewise the difference between the mean stress and pore pressure will be referred to as the "effective mean stress". A variety of porous rocks will be reviewed, and for reference we compile in Table 1 their petrophysical data.

Under relatively small stress increments, the mechanical response of a porous rock can be described as poroelastic, although it is often nonlinear especially if reversible closure and opening of microcracks and pores are involved (Zimmerman, 1991; Guéguen and Palciauskas, 1994; Mavko et al., 1998). The linear and nonlinear elastic behavior of porous rock was reviewed by Wong

Table 1
Petrophysical description of the rocks investigated in this study.

Rock name	Porosity, %	Grain radius, mm	Composition	Reference
Darley Dale sandstone	13	0.17	Quartz: 66%, feldspar 21%, mica 3%, clay ~6%	Baud et al. (2000b) Baud et al. (2006)
Rothbach sandstone	19.9	0.23	Quartz: 68%, feldspar 16%, oxides and mica 3%, clay ~6%	Wong et al. (1997) Louis et al. (2009)
Kayenta sandstone	21	0.15	Quartz: 81%, feldspar 16%, carbonate 1%	Wong et al. (1997)
Berea sandstone	21	0.13	Quartz: 75%, feldspar 10%, carbonate 5%, clay 10%	Zhang et al. (1990b) Baud et al. (2000b)
Adamswiller sandstone	22.6	0.09	Quartz: 71%, feldspar 9%, oxides and mica 5%, clay ~11%	Wong et al. (1997) Baud et al. (2006)
Bentheim sandstone	22.8	0.105	Quartz: 95%, 3% kaolinite, 2% orthoclase	Klein et al. (2001) Baud et al. (2004)
Diemelstadt sandstone	24.3	0.08	Quartz: 68%, feldspar 26%, oxides: 4%, mica 2%	Tembe et al. (2008)
Bleurwiller sandstone	25.2	0.112	Quartz: 50%, feldspar 30%, oxides: and mica 2%	Fortin et al. (2005) Tembe et al. (2008)
Saint Peter sandstone	29	0.14	Quartz: 99%	
Boise sandstone	35	0.28	Quartz: 67%, feldspar 14%, mica 2%, clay ~13%	Wong et al. (1997) Baud et al. (2000b)
Solnhofen limestone	3	0.005	~100% calcite	Baud et al. (2000a)
Tavel limestone	10.4	0.005	~100% calcite	Vajdova et al. (2004a) Vajdova et al. (2010)
Indiana limestone	13–19	0.005 0.3	~100% calcite	Vajdova et al. (2004a) Zhu et al. (2010)
Majella limestone	30	0.027	~100% calcite	Baud et al. (2009) Zhu (2010)
Tuffo del Palatino	32	–	Leucite, clinopyroxene, yellow scoria clasts,	Zhu et al. (2011)

et al. (2004). If the deformation involves significant irreversible strain, then it is unlikely that a poroelastic model can capture adequately the mechanical responses. Laboratory studies of porous rocks have demonstrated that the inelastic and failure behaviors have at least four common attributes. [1] Under relatively low confinement, the inelastic deformation is dilatant, and after attaining a peak stress, the sample would fail by the development of one or more shear bands accompanied by strain softening. [2] Inelastic deformation in the form of volume decrease may develop even in the absence of any deviatoric stresses. Significant compaction can occur in response to purely hydrostatic loading, due to an increase in confining pressure or decrease in pore pressure. [3] Under relatively high confinement, a deviatoric stress field promotes the onset and development of inelastic compaction. If we compare the volumetric strains in two systems subjected to identical effective mean stress, compaction in a nonhydrostatically loaded system is enhanced relative to that in a hydrostatically loaded system. This phenomenon is referred to as "shear-enhanced compaction" (Curran and Carroll, 1979). [4] While shear localization is often inhibited by compactancy (and promoted by dilatancy), recent studies have shown that several modes of strain localization can actually develop in a compactant rock.

2.1.1. Hydrostatic compaction

Fig. 1a shows the hydrostatic compaction behavior for seven saturated sandstones with porosities ranging from 20% to 35% (Zhang et al., 1990a; David et al., 1994; Wong et al., 1997; Baud et al., 2006). Typically, the mechanical response of a porous sandstone is nonlinear even in the elastic regime. Under hydrostatic loading, the pore space is initially tightened by elastic deformation, and as a result, it becomes progressively stiffer, as manifested by a decrease in the compressibility. However, as the hydrostatic loading increases, the sample reaches a point where it suddenly becomes more compliant, showing a dramatic increase in compaction. If a sample is loaded beyond this inflection point and then unloaded, the permanent compaction is significant implying the deformation is inelastic (Brace, 1978). This inflection point P^* occurs at a wide range of effective pressures, from 80 MPa to 420 MPa for the seven sandstones in Fig. 1a. For a silicilastic rock (and unconsolidated aggregates), the magnitude of P^* is primarily controlled by the porosity and average grain size (Zhang et al., 1990a). It was also emphasized by Wong et al. (1997) that other factors (such as phyllosilicate content, cementation and bedding) can appreciably influence the critical pressure P^*, such that at the same porosity and grain size the critical pressure P^* may deviate by as much as a factor of two, as illustrated by the data of Tembe et al. (2008) for Diemelstadt, Bleurswiller, and Bentheim sandstones. In the presence of water, the onset of inelastic compaction typically occurs at a lower critical stress such that more inelastic compaction would accumulate at the same stress level (Baud et al., 2000b; Tembe et al., 2008). This contrast is illustrated by the hydrostats of nominally dry and water-saturated samples of Dielmestadt sandstone in Fig. 1b.

The hydrostatic compression behavior in a porous carbonate rock such as chalk (Azeemudin et al., 1994; Risnes et al., 1996; Homand and Shao, 2000), limestone (Mowar et al., 1994; Fabre and Gustkiewicz, 1997; Baud et al., 2000a; Vajdova et al., 2004a; Dautriat et al., 2011; Xie et al., 2011) and tuffs (Lockner and Morrow, 2008; Zhu et al., 2011) is qualitatively similar. We compile in Fig. 2 the critical pressures P^* for pore collapse that have been measured for limestones and tuffs. There is an overall trend for the critical pressure to decrease with increasing porosity: P^* decreases significantly in the porosity range of 3–15%, followed by a more gradual decrease in the range of 15–45%. The data selected are all for nominally dry or oil-saturated samples. It is likely that the water-weakening effect in a carbonate rock is comparable to or maybe exceeds that in a siliciclastic rock, which is to be confirmed by more systematic investigations.

2.1.2. Brittle faulting and compactive cataclastic flow

The mechanical response to isotropic (hydrostatic) loading and to anisotropic loading is fundamentally different. While hydrostatic loading always gives rise to compaction, whether porosity decreases or increases in response to a deviatoric stress field depends on the trade-off between pore collapse mechanisms and the opposite tendency for microcracking to dilate the sample. In porous carbonate rocks, the cataclastic flow regime typically involves two distinct stages, an initially compactant one that

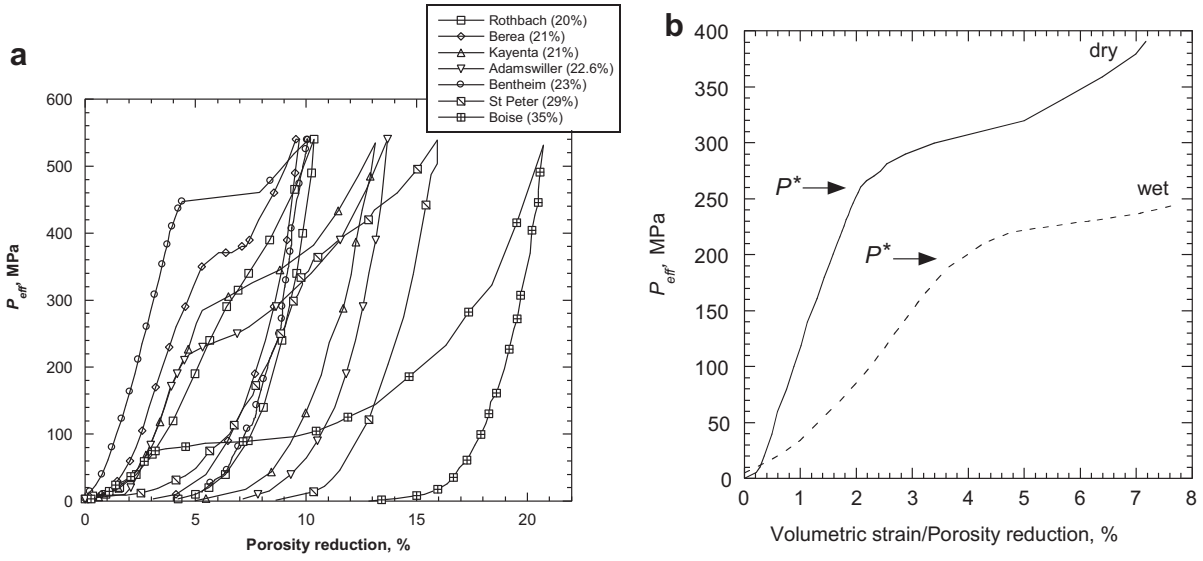


Fig. 1. (a) Hydrostats of seven sandstones. Each of the hydrostats show a sharp inflection point which corresponds to the critical effective pressure P^* for the onset of grain crushing and pore collapse. The experiments were conducted on samples saturated with distilled water and under drained conditions. The nominal porosity of each sandstone is also indicated in bracket. (b) Comparison between the dry and wet hydrostats for Diemelstadt sandstone (Tembe et al., 2008). The critical pressure P^* is indicated for both cases.

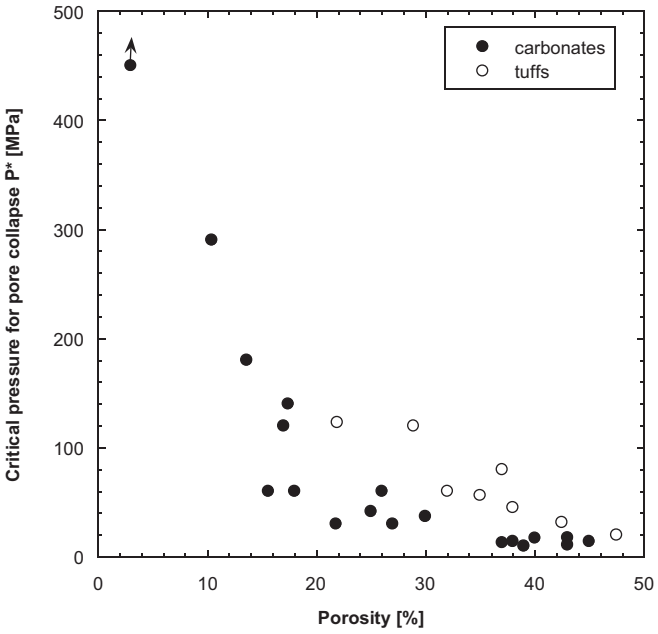


Fig. 2. Compilation of pore collapse pressure for carbonate rocks (close circles) and tuffs (open circles).

evolves to become dilatant. Distributed cataclastic flow may persist over large strains if the multiplicity of microcracks and pores do not coalesce to result in strain localization. However, recent laboratory studies have shown that the brittle-ductile transition in porous sandstone often involves an intermediate failure mode between the two end-members of brittle faulting and compactive cataclastic flow: strain localization may develop while the sample is undergoing inelastic compaction, corresponding to an intermediate regime of compaction localization.

Fig. 3a and b show a set of mechanical data representative of the brittle-ductile transition in a porous limestone. Nominally dry samples of Tavel limestone were deformed in conventional triaxial compression at confining pressures ranging from 10 MPa to 240 MPa (Vajdova et al., 2004a). Data for differential stress $\sigma_1-\sigma_3$ versus axial strain for eight selected experiments are shown in Fig. 3a. At confining pressures of 10, 20 and 30 MPa the mechanical response and failure mode are typical of the brittle faulting regime. The differential stress attained a peak, beyond which strain softening was observed. The peak stress shows a positive correlation with the confining pressure. Visual inspection of the failed samples revealed the development of localized failure subparallel to the sample axis at 10 MPa pressure, while shear bands inclined to the axis were observed in the samples faulted at higher confining pressures. At confining pressures of 100 MPa and higher, the initial mechanical responses are typical of the cataclastic flow regime. The slopes of the differential stress-axial strain curve were positive, and strain localization was not observed in the failed specimens.

As noted earlier, additional insights can be gained by analyzing the mean stress $(\sigma_1 + 2\sigma_3)/3$ versus volumetric strain and contrasting the mechanical responses under hydrostatic and non-hydrostatic loading. In a conventional triaxial compression experiment, the nonhydrostatic and hydrostatic loadings are coupled, in that if the axial stress increases by an increment $\Delta\sigma_1$ while the confining pressure is maintained constant, then the mean stress and differential stress would increase by the amounts $\Delta\sigma_1/3$ and $\Delta\sigma_1$, respectively. In a poroelastic material, the volumetric strain should be controlled solely by the mean stress, and therefore in a plot (such as Fig. 3b) relating the two the triaxial data (solid curves) should coincide with the hydrostat (dashed curves) as long as the deformation is poroelastic. Deviations of the triaxial curves from the hydrostat would imply that the volumetric strain is dependent on not only the mean stress, but also the deviatoric stresses, with the implication that the material has yielded and is no longer poroelastic. It can be seen in Fig. 3b that in the cataclastic flow regime (at confining pressures of 100 MPa and higher), the triaxial curve for a given confining pressure would coincide with the hydrostat up to a critical stress state (indicated by C^* in Fig. 3b), beyond which there was an accelerated decrease in volume in comparison to the hydrostat. At stress levels beyond C^*, the deviatoric stress field can significantly increase the compactive strain, a phenomenon that Curran and Carroll (1979) referred to as "shear-enhanced compaction". In contrast, the volumetric change at confining pressures of 30 MPa and below was such that the compaction decelerated in comparison with the hydrostat beyond critical stress states marked as C', and this corresponds to the onset of dilatancy. Accordingly these two critical stress states (C^* and C') mark the termination of the poroelastic regime and the onset of inelastic yield.

A more complex failure mode was observed at 50 MPa of confinement. A switch from shear-enhanced compaction to dilatancy was observed while the sample strain hardened. After attaining a peak stress, the sample strain softened and developed conjugate shear bands visible on the surface. It can also be seen in Fig. 3b that in the cataclastic flow regime (at confining pressures of 100 MPa and higher), after they had undergone certain amount of strain hardening the samples consistently switched from compaction to a third stage of dilation. That cataclastic flow in a porous rock would evolve from transient compactancy to culminate in dilatancy is a phenomenon which has been observed in limestones with porosities ranging from 3% up to 30% (Baud et al., 2000a; Vajdova et al., 2004a; Vajdova et al., 2012; Dautriat et al., 2011). As indicated for the experiment at 100 MPa confining pressure in Fig. 3b, we denote this critical stress state at the transition from compactant to dilatant flow by $C^{*\prime}$. While the differential stress C^* at the onset of shear-enhanced compaction shows a negative pressure dependence, the differential stress at $C^{*\prime}$ shows a positive correlation with mean stress and confining pressure, following a trend similar to that for the stress C' associated with the onset of dilatancy under low confinement. It is plausible that shear localization may develop if such samples were to be deformed beyond $C^{*\prime}$ to larger strains.

A similar methodology can be adopted for analyzing the mechanical data of a saturated rock sample subject to a constant pore pressure, but instead of the confining pressure and mean stress, one should subtract the pore pressure from these two quantities and consider the effective pressure and effective mean stress. This is illustrated in Fig. 3c by the mechanical data of Zhu et al. (2011) for porous tuff acquired at a fixed pore pressure of 10 MPa. The dashed curve is for hydrostatic compression of a water-saturated sample of Tufo del Palatino up to effective pressure of 110 MPa (confining pressure of 120 MPa), showing a critical effective pressure P^* of 41 MPa. Shear-enhanced compaction and cataclastic flow were observed in the three samples triaxially compressed at effective pressures ranging from 10 MPa to 30 MPa. The critical stress C^* at the onset of shear-enhanced compaction for the tuff sample deformed at effective pressure of 30 MPa is marked in Fig. 3c. A transition to dilatant flow was not observed in these experiments on tuff. The sample deformed at effective pressure of 5 MPa showed a transitional behavior, with small stress drop accompanied by shear-enhanced compaction.

It should be noted that there are other approaches that one can take to identify the yield stress from experimental data. For example, one may conduct unloading–reloading cycles repeatedly

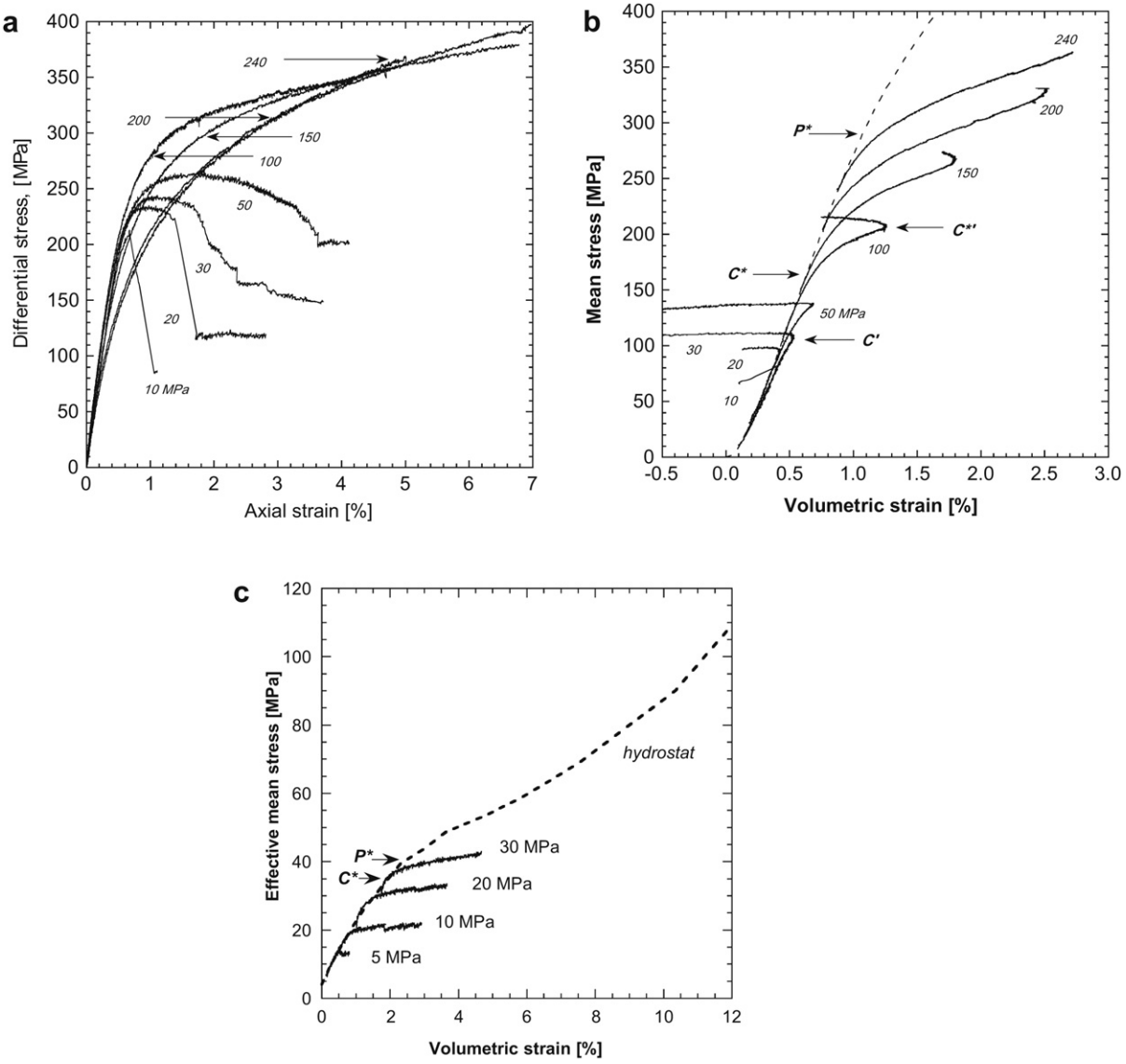


Fig. 3. Representative mechanical data for selected triaxial compression experiments on Tavel limestone. (a) Differential stress is plotted versus axial strain. (b) Volumetric strain is plotted versus mean stress, and for reference the hydrostatic data are shown as the dashed line. Numbers next to each curve indicate the confining pressure maintained during the experiments. Arrows in Fig. 3b mark critical stress states (onset of dilatancy C' in experiment at confining pressure of 30 MPa, onset of shear enhanced compaction C^* and transition from shear-enhanced compaction to dilatancy $C^{*\prime}$ in experiment at confining pressure of 100 MPa). (c) Volumetric strain versus mean stress for triaxial compression experiments on wet Alban Hill tuff (Tufo del Palatino). For reference, the hydrostatic data are shown as the dashed curve. Numbers next to each curve indicate the effective pressures maintained during the experiments. The critical pore collapse pressure P^* and the onset of shear-enhanced compaction C^* at effective pressure of 30 MPa are marked by arrows.

to pinpoint the stress level for the onset of inelastic behavior. While such an approach can provide results that are mechanically consistent, it is somewhat tedious in comparison to the methodology we just described. Of course, the usefulness of this methodology hinges on the availability of samples from a block of rock with reproducible mechanical behavior. This is not viable, for instance, in soil mechanics. The methodology also requires the availability of a pressure vessel that can go to relatively high pressures, so that the hydrostatic yielding behavior beyond the critical pressure P^* can be observed. Some studies attempted to associate the yield stress with a point in the stress–strain curve that shows a drastic change in slope (e.g., Cuss et al., 2003; Dautriat et al., 2011). Given the intrinsic nonlinearity of stress–strain curves of porous rocks, this type of approach involves certain degree of ambiguity. Tembe et al. (2007) proposed the use of a "modified undrained" experiment, in which an isoporosity contour is mapped out while manipulating the confining pressure and axial stress and maintaining a constant pore pressure. They demonstrated that the isoporosity contour would correspond to the initial yield cap if the elastic strain is negligible, which is the case when relatively low differential stresses are involved when a porous rock yields under high confinement.

2.1.3. Development of compaction localization

Bentheim sandstone is a porous rock that fails by the development of compaction bands over an unusually broad range of pressure conditions. Here we present the mechanical data for eleven water-saturated samples of Bentheim sandstone deformed at effective pressures ranging from 10 MPa to 350 MPa (Baud et al., 2006). The data for differential stress as a function of axial strain in five selected experiments are shown in Fig. 4a, and data for the effective mean stress as a function of porosity reduction in all eleven experiments are shown in Fig. 4b. For reference the

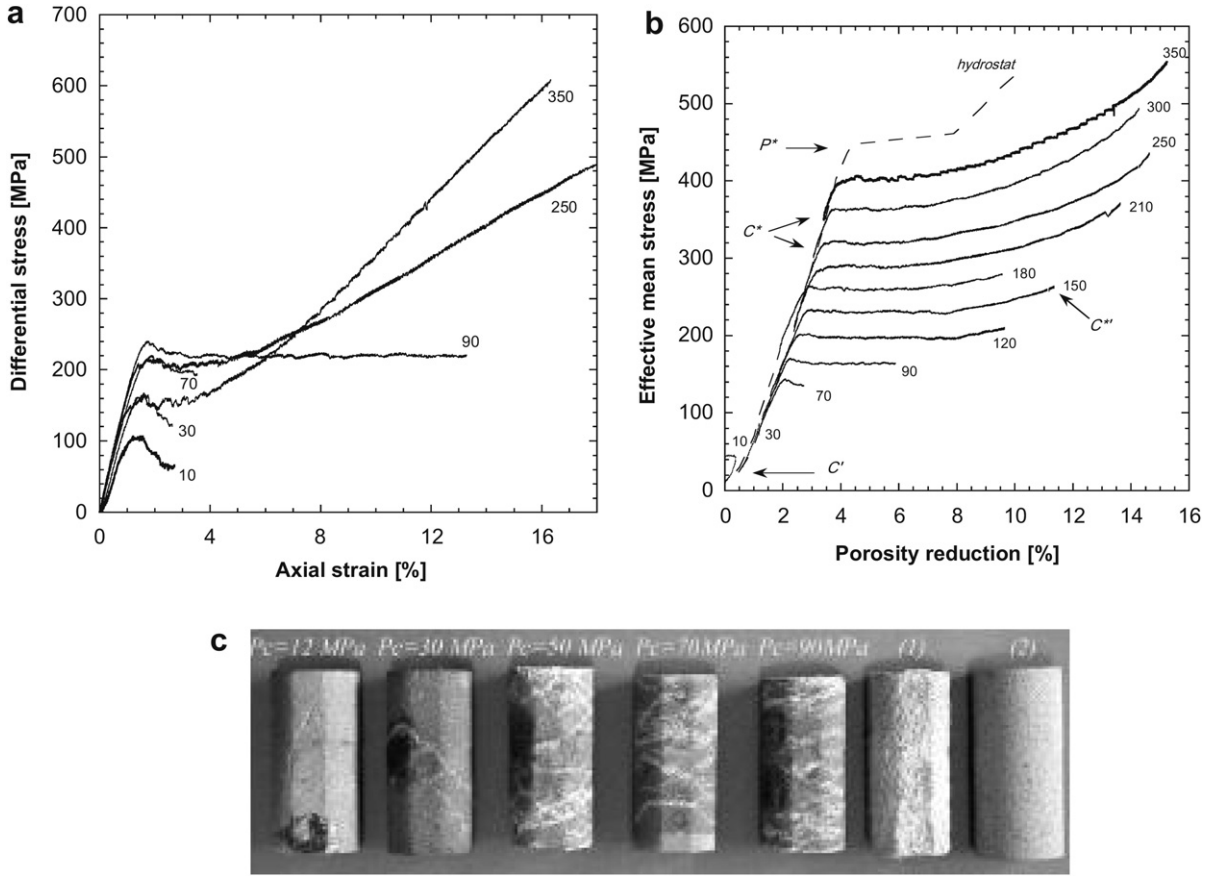


Fig. 4. Representative mechanical data for selected triaxial compression experiments on Bentheim sandstone. (a) Differential stress is plotted versus axial strain. (b) Effective mean stress as a function of porosity reduction for triaxial compression experiments at effective pressures indicated by numbers (in MPa) next to each curve. For reference the hydrostatic data are shown as the dashed line. Arrows mark critical stress states C', C^* and $C^{*\prime}$ for selected experiments and the critical pressure P^* for the hydrostatic experiment. (c) Series of photographs illustrating the failure modes of Bleurswiller sandstone samples deformed at different confining pressures (Guéguen and Fortin, 2005). The diameter of the initially intact sample (2) was 40 mm and the length was 80 mm. Sample (1) was deformed under hydrostatic condition.

hydrostat is also shown as the dashed curve in Fig. 4b. Brittle faulting (accompanied by dilatancy and strain softening) was observed for the two samples deformed at effective pressures of 10 and 30 MPa. As can be seen from Fig. 4b, shear-enhanced compaction was observed in the samples deformed at effective pressures of 70 MPa and higher.

The failure modes of these saturated samples (and dry samples deformed at similar pressure conditions) have been described by Klein et al. (2001), Wong et al. (2001), Baud et al. (2004) and Baud et al. (2006). Using the terminology introduced by Baud et al. (2004) on the basis of their data for five sandstones with porosities ranging from 13% to 24%, a localized structure that has a thickness of only a few (say $\leq$3) grains is referred to as a "discrete band", whereas thicker structures are called "diffuse band". Discrete compaction bands were observed in the failed samples at a broad range of effective pressures from 120 to 350 MPa. For samples that failed in the transitional regime (between brittle faulting and compaction banding), high-angle shear bands (and a few discrete bands) were observed at effective pressures of 70 and 90 MPa.

To our knowledge, this mode of compaction localization has been observed in only three sandstones. The development of compaction bands similar to those in Bentheim sandstone has been reported in Bleurswiller sandstone (Fortin et al., 2007; Tembe et al., 2008) and Diemelstadt sandstone (Baud et al., 2004; Louis et al., 2006; Tembe et al., 2008) under both dry and saturated conditions. Fig. 4c shows the failure modes manifested by the exterior appearances of five triaxially compressed samples and a hydrostatically compressed sample of Bleurswiller sandstone deformed by Guéguen and Fortin (2005).

In most other sandstones, discrete compaction bands are not observed and distributed cataclastic flow is the failure mode under high confinement, although high-angle shear bands and diffuse compaction bands may develop in the transitional regime between brittle faulting and cataclastic flow (Olsson and Holcomb, 2000; Bésuelle et al., 2000, 2003; Baud et al., 2004). Unlike limestone for which the phenomenon of transient compactancy evolving to dilatancy (Fig. 3b) is widely observed, similar behavior has been observed in only a few sandstones, possibly because the transition to dilatancy in a siliciclastic rock requires significantly larger strains and higher stresses (Baud et al., 2006). The first observation was probably by Schock et al. (1973) in Lance sandstone (with 7.5% porosity). In Fig. 4b the four Bentheim sandstone samples deformed at effective pressures between 120 and 210 MPa were strained to the critical stress level $C^{*\prime}$, at the onset of dilatant cataclastic flow. In addition, Baud et al. (2006) also tabulated two sets of $C^{*\prime}$ data for Darley Dale and Bentheim sandstones.

2.2. Brittle strength envelope and compactive yield cap

The mechanical data for porous rock indicate that the brittle strength and compactive yield stress have fundamentally different

dependence on pressure, and this can be highlighted by mapping the critical stresses in the stress space. In Figs. 5 and 6 we have compiled the critical stress data for sandstone, limestone and tuff in the stress space, with coordinates given by the effective mean stress $P(=(\sigma_1+2\sigma_3)/3-p_p)$ and the differential stress $Q(=\sigma_1-\sigma_3)$. The choice of P and Q has the advantage that these two variables are directly related to the first and second invariants of the stress tensor, respectively. In the brittle regime (open symbols), there is a positive correlation between the peak differential stress and effective mean stress (Fig. 5a, b, Fig. 6a, b), as expected for Mohr-Coulomb failure.

In the cataclastic flow regime, we have plotted the stress at the onset of shear-enhanced compaction. The C^* data (solid symbols) map out the envelopes (caps) for the initiation of compactive yield. Overall there is an inverse correlation between P and Q at the critical stress states at the onset of shear-enhanced compaction (Fig. 5a, b, Fig. 6a, b). The brittle–ductile transition can be identified with the stress state at the intersection of this compactive yield cap with the Mohr-Coulomb envelope for brittle faulting.

The data for sandstone (Fig. 5) with initial porosities ranging from 13 to 35% underscore that shear-enhanced compaction and cataclastic flow can occur over a broad range of stress conditions (Wong et al., 1997; Baud et al., 2006). The compactive yield cap of Berea sandstone has stress values more than five times greater than those of Boise sandstone. Since the former has a porosity of 21% and the latter a porosity of 35%, this indicates that porosity exerts important influence over the compactive yield behavior. However, other microstructural parameters (in particular grain size, as well as cementation, and clay content) may also be important since there is considerable discrepancy between the yield envelopes of Berea and Kayenta sandstones, both of which have porosity of 21%. The brittle strength and compactive yield stress data for limestone (Baud et al., 2000a; Vajdova et al., 2004a; Bemer et al., 2004; Baud et al., 2009; Dautriat et al., 2011; Xie et al., 2011) and tuff (Zhu et al., 2011) show similar contrast in their sensitivity to pressure (effective mean stress). There is an overall trend for the compactive yield cap to expand with decreasing porosity.

Since one end of the compactive yield envelope is "anchored" at a location on the P-axis corresponding to P^*, this implies that if a porous rock requires a relatively high effective pressure for pore collapse to occur, then the critical stress states C^* for the onset of shear-enhanced compaction are also expected to be high. In other words, the critical pressure P^* acts as a scaling parameter for the magnitudes of the compactive yield stresses C^*, and in this sense the critical pressure P^* provides a quantitative measure of the "brittleness" of a porous rock (Wong et al., 1997).

Wong et al. (1997) have observed that their sandstone data can be described by an elliptical cap in the P–Q stress space:

$$\frac{(P/P^*-\gamma)^2}{(1-\gamma)^2}+\frac{(Q/P^*)^2}{\delta^2}=1 \qquad (1)$$

with $\gamma\approx 0.5$ and δ between 0.5 and 0.7. Accordingly to the Hertzian fracture model of Zhang et al. (1990a), porosity Φ and grain size R exert control over the critical pressure P^* for the onset of grain crushing process obeying a power law $P^*\propto(\Phi R)^n$, with $n=-1.5$. That the compactive yield stress data are bracketed by the normalized caps in turn implies that the two microstructural parameters have similar influence on the compactive yield envelope, in that the initial yield cap for a sandstone is expected to expand with decreasing porosity and grain size following the power law. Laboratory data for siliciclastic rock, sand and glass sphere fall on an overall trend defined by the power law, with an exponent ≈ -1.5 that varies somewhat among different types of geomaterials (Wong et al., 1997). It should be noted that other factors (including mineralogy, presence of water and bedding) can appreciably influence the grain crushing pressure, such that at the same porosity and grain size the yield stresses may deviate by as much as a factor of 2 from the trend predicted by the Hertzian fracture model (Baud et al., 2000b; Tembe et al., 2008).

When normalized by the critical pressure P^*, the P and Q data for limestones (Zhu et al., 2010) and tuffs (Zhu et al., 2011) at the critical stress state C^* for the onset of shear-enhanced compaction follow a very similar trend, as illustrated in Fig. 6c. An elliptical cap can fit the normalized data of limestone and tuff, but in comparison

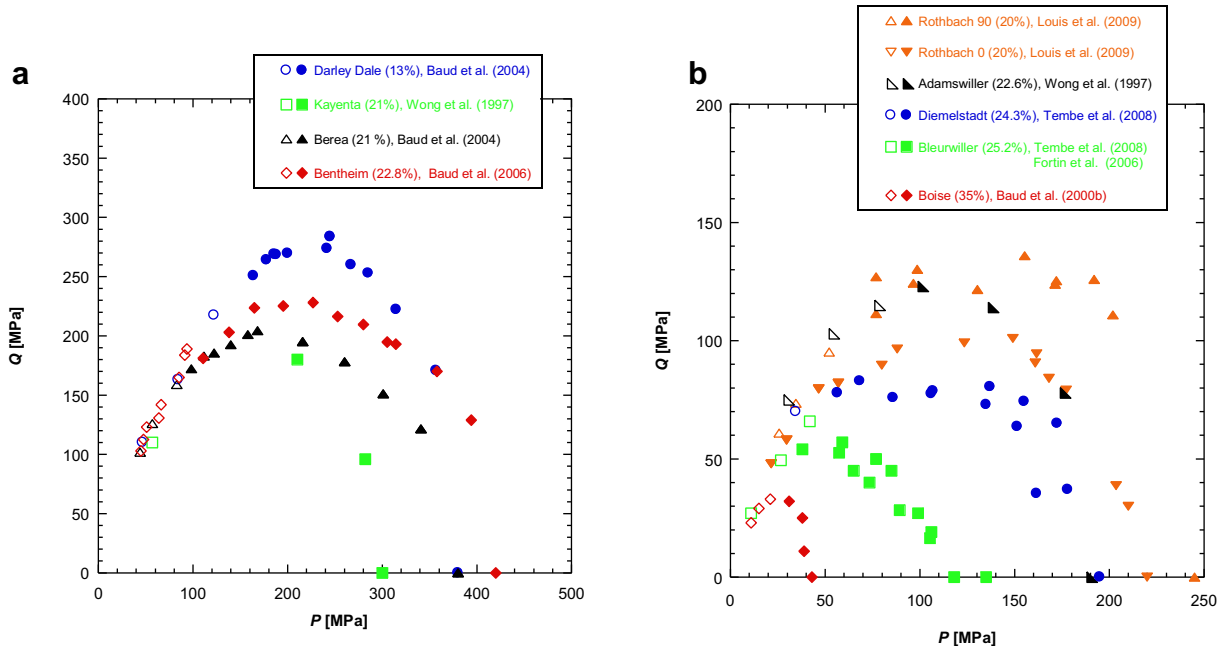


Fig. 5. Compilation of peak stress and critical stress state C^* data shown in the stress space for 8 sandstones with porosity ranging from (a) 13%–23% and (b) 20–35%.

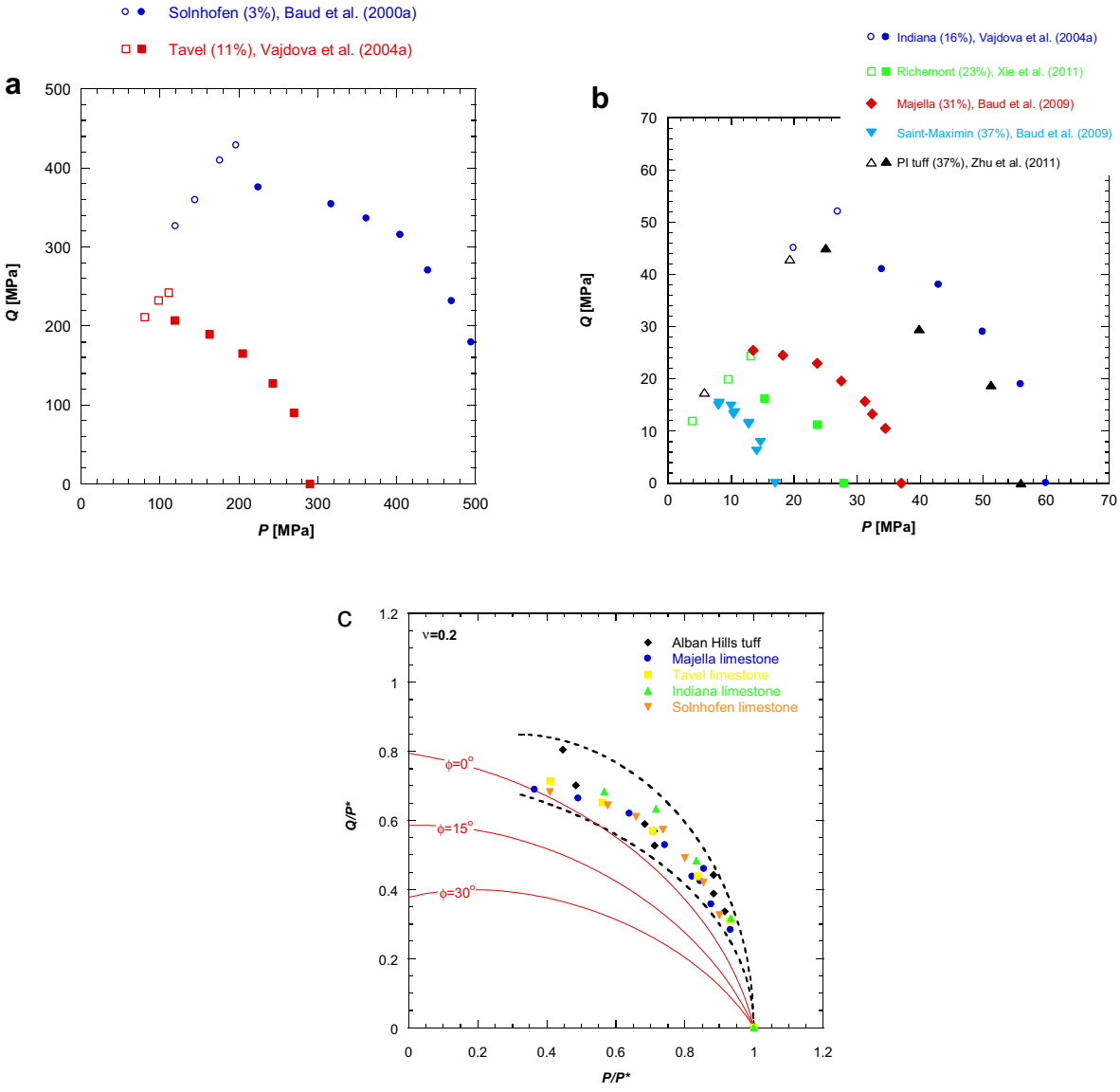


Fig. 6. Peak stress and stress state C^* are shown in the stress space. Differential stress (Q) as a function of mean stress (P), for (a) Solnhofen and Tavel limestones, (b) Indiana, Richemont, Majella and Saint-Maximin limestones, and Alban Hills Tuff. (c) Onset of shear-enhanced compaction C^* for the data of Fig. 6a and b normalized by the pore collapse pressure P^*.

to the sandstone cap, it has lower value of γ ($\leq$0.4) and higher value of δ ($\geq$0.7). As noted in the figure, the normalized caps for dry and water-saturated samples of tuff follow very similar trend. A similar conclusion also applies to data for dry and water-saturated samples of sandstone (Baud et al., 2000b; Baud et al., 2006). Since the critical pressure P^* of a nominally dry sample is higher than that in a water-saturated one, they have caps of similar elliptical shape but at different stress levels.

Compaction data for sand (Karner et al., 2003, 2005) and calcite-quartz aggregates (Xiao and Evans, 2003) at elevated temperatures also indicate that the cap would shrink with increasing temperature for a given strain rate. In a recent study, Rutter et al. (2009) deformed serpentinite samples with porosity that had developed by the dehydration of lizardite to olivine plus talc. Their compactive yield data at elevated temperatures map out caps qualitatively similar to those of carbonate and siliciclastic rocks. Their data indicate that porosity cannot be the only microstructural attribute that controls the compactive yield behavior, since the yield stresses of samples that had undergone dehydration under different effective pressures vary by as much as a factor of two, even though the porosities induced by hydration were almost identical (~20%).

2.3. Evolution of compactive yield cap with strain hardening

The evolution of strain hardening can be visualized in the stress space by mapping out the stresses as a function of an appropriate hardening parameter. An example is given in Fig. 7 for Bentheim sandstone using the plastic volumetric strain ε_v^p as hardening parameter (Baud et al., 2006). Since in a sandstone plastic deformation of the solid grains is negligible at room temperature under the pressure conditions in the experiments, the plastic (irreversible) porosity change $\Delta\Phi^p$ represents the bulk of the plastic volumetric strain, and to a first approximation we can take $\varepsilon_v^p = \Delta\Phi^p$. From a plot of the total porosity change versus effective mean stress (Fig. 4b) the plastic porosity change $\Delta\Phi^p$ at a given point beyond C^* was evaluated by subtracting from the total porosity change $\Delta\Phi$ the

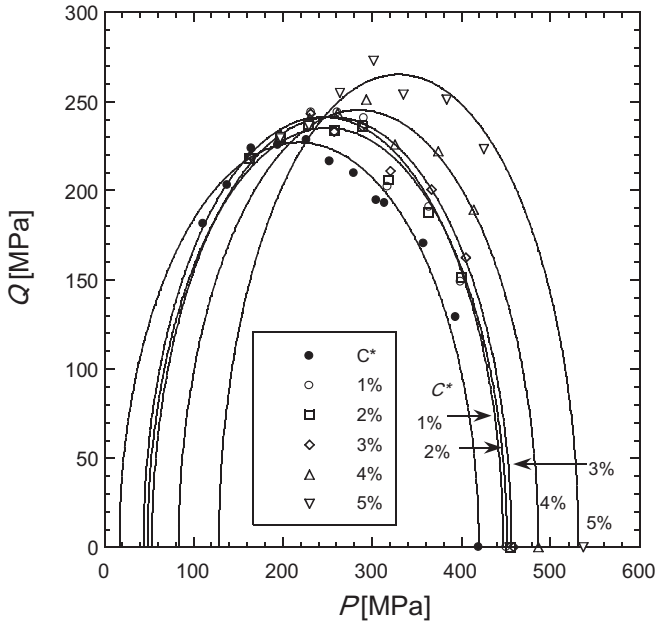


Fig. 7. Comparison between DiMaggio and Sandler's (1971) cap model and experimental data on the initial yield stress C^* and evolution of the yield stress as a function of plastic volumetric strain for Bentheim sandstone.

elastic porosity change which is assumed to be given by the linear portion during initial loading.

As the sandstone samples strain hardened, the yield cap maintained an elliptical shape and its center progressively shifted to higher P values with the accumulation of plastic volumetric strain. Baud et al. (2006) systematically analyzed such data for four porous sandstones, and compared them with the predictions of constitutive models discussed in the next section. There is a paucity of similar studies on other porous rocks such as limestone and tuff, and it is desirable to conduct systematic investigations in the future.

3. Constitutive models and bifurcation analysis of strain localization

Since constitutive models for the development of dilatant failure in the brittle faulting regime have been reviewed extensively (Paterson and Wong, 2005; Jaeger et al., 2007), our focus here will be on constitutive modeling of the ductile behavior of a porous rock. Plasticity theory provides a constitutive framework for the analysis of inelastic yield and compaction in a granular material such as soil, porous sandstone, diatomites and carbonate rocks (Chen, 1984; Desai and Siriwardane, 1984; Fossum and Fredrich, 2000; Davis and Selvadurai, 2002; Grueschow and Rudnicki, 2005). In particular two classes of plasticity models are often invoked: the critical state model (Schofield and Wroth, 1968) which has had a profound impact in soil mechanics, and the cap model which was formulated by DiMaggio and Sandler (1971) with both soil and porous rock in mind.

In the 20th Bjerrum Lecture, Muir Wood (2007) presented a succinct summary and critical review of the theory of critical state soil mechanics. He noted that "critical state soil mechanics in its weak form simply implies a recognition that many aspects of soil behavior can be better understood if they are related to density changes so that stress states and density are considered together". Accordingly density (or porosity) change is the only microstructural attribute that enters into the theory, and its most successful application is probably the Cam clay model, which can capture the constitutive behavior of clay using as few as five parameters (Muir Wood, 1990; Davis and Selvadurai, 2002; Guéguen et al., 2004).

Recent studies have shown that the application of this theory to sand is not as straightforward as clay, especially in the presence of grain crushing. "In its stronger form critical state soil mechanics is about critical states. What happens when we get to large strains at the end of a shearing test? Are there asymptotically approached limiting states in which shearing can continue indefinitely with no further changes of state" (Muir Wood, 2007). There are sand deformation data which indicate that the predicted critical state line would actually emerge more as a "fuzzy" critical state zone, and to be compatible with the data, the theory must be modified by, for example, incorporating additional microstructural parameter such as a "grading state index" related to particle size distribution (Muir Wood and Maeda, 2008). To address this issue, Einav (2007a, 2007b) went so far as to propose foregoing the critical state framework and formulate alternatively a "breakage mechanics", which he has also attempted to apply to porous sandstone (Nguyen and Einav, 2009; Das et al., 2011).

The key concept that microstructural parameters (such as grain size) must be considered in addition to porosity in modeling the cataclastic failure of rock has been built into rock mechanics literature, long before it began to be recognized in soil mechanics. Furthermore the experimental approach by which the initial yield stress is identified with the onset of shear-enhanced compaction by comparing hydrostatic and nonhydrostatic loadings is quite unique to rock mechanics, in that a similar methodology is simply not viable in soil mechanics which requires quite complicated tracking of the highly variable porosity for inference of the inelastic and yield behavior. Given these limitations intrinsic to the theory and certain fundamental differences between soil and rock, we consider it inappropriate and somewhat misleading to refer to "critical state soil mechanics" as an abstract entity or established theory that can be applied directly to porous rock deformation or even "unify" it (e.g., Cuss et al., 2003; Schultz and Siddharthan, 2005; Sheldon et al., 2006). A more useful approach is to consider critical state models as a class of plasticity models formulated for porous materials with the common premise that there exists a critical state line, which may depend not only on porosity (as in Cam clay) but also some other microstructural attributes. The yield envelope is not necessarily circular in the P-Q space (as originally stipulated in the Cam clay model), and under hydrostatic loading the change of porosity (or void ratio) may also have a more complex dependence on the effective pressure than that specified for Cam clay.

When a critical state model is developed for analyzing porous rock deformation, the constitutive model should be such that its predictions are in basic agreement with the experimental data. Following Baud et al. (2006), we will here focus on a specific critical state model that was proposed by Carroll (1991) for porous reservoir rocks. We will compare it with the cap model, which is another class of plasticity models also with many different formulations (e.g., Desai and Siriwardane, 1984; Lubarda et al., 1996; Fossum and Fredrich, 2000). Here we will consider one version that is basically the cap model first proposed by DiMaggio and Sandler (1971).

3.1. Two plasticity models with elliptical yield caps

The particular critical state and cap models we consider here have four common features. [1] They both assume that the plastic yield behavior of the compactant rock is isotropic, so that the plastic yield function depends only on the stress invariants. [2] The yield function is assumed to be independent of the third stress invariant. [3] The strain hardening behavior is characterized by a yield function that depends

solely on the plastic volumetric strain. [4] Both models assume the plastic flow to be associative satisfying the normality condition. The first step in developing a plasticity model is to identify an appropriate yield function f in the stress (σ_{ij}) space. If the yield behavior is isotropic, then the yield behavior can be described by a function dependent only on the three stress invariants, and furthermore if the yield behavior can be considered to be independent of the third stress invariant, then f is a function of only the first and second stress invariants, $I_1 = (\sigma_{11} - p_p) + (\sigma_{22} - p_p) + (\sigma_{33} - p_p)$ and $J_2 = [(\sigma_{11} - \sigma_{22})^2 + (\sigma_{22} - \sigma_{33})^2 + (\sigma_{33} - \sigma_{11})^2]/6 + \sigma_{12}^2 + \sigma_{23}^2 + \sigma_{13}^2$. The two constitutive models considered here both adopt a yield function of the form

$$f(\sigma_{ij}) = \frac{(I_1 - c)^2}{a^2} + \frac{J^2}{b^2} - 1 = 0 \tag{2}$$

which corresponds to an ellipse centered at (c, 0) with major and minor semi-axes of a and b. The three parameters for the ellipse can be evaluated from conventional triaxial tests on saturated samples by noting that under axisymmetric loading the first stress invariant is related to the effective mean stress P by $I_1 = 3P$ and the second invariant to the differential stress Q by $3J_2 = Q^2$. Accordingly the constitutive model prescribes the triaxial compression data on yield stress to fall on the ellipse with semi-axes $A = a/3$ and $B = 3b$ and center at ($C = c/3$,0):

$$\frac{(P - C)^2}{A^2} + \frac{Q^2}{B^2} = 1 \tag{3}$$

This choice of an elliptical cap is motivated by the laboratory data for the initial yield stresses of sandstones, limestones and tuffs at the onset of shear-enhanced compaction, which we reviewed in an earlier section. With reference to the elliptical normalized cap associated with Equation (1), the parameters are related by $\gamma = c_o/(3P^*) = C_o/P^*$ and $\delta = b_o/(3P^*) = B_o/P^*$. As the rock undergoes inelastic deformation and work hardening, the yield function f (and therefore the three parameters a, b and c) are assumed to depend only on the plastic volumetric strain ε_v^p. The initial yield stresses (corresponding to C^* at the onset of shear-enhanced compaction) are expected to fall on the initial elliptical cap for $\varepsilon_v^p = 0$. We will use A_o, B_o and C_o to denote the semi-axes and center of this initial cap.

3.1.1. Carroll's critical state model

In Carroll's (1991) model, evolution of the yield envelope as a function of plastic volumetric strain during work hardening can be illustrated by the schematic diagram in Fig. 8a. The three parameters for the ellipse are subjected to two constraints. [1] One of the semi-axis remains constant so that $A = A_0$ even if ε_v^p becomes greater than zero. [2] The other two parameters are related to one another, in that they are required to satisfy

$$B = B_0 + m(C - C_0) \tag{4a}$$

As illustrated in Fig. 8a this condition imposes the rather stringent constraint that the yield cap will always evolve with its tip located at (C, B) moving along the "critical state line" with slope m, or equivalently the coordinates (c, b) always fall on the line

$$b = b_0 + \left(m/\sqrt{3}\right)(c - c_0) \tag{4b}$$

As emphasized by Schofield and Wroth (1968), this second condition corresponds to a "kernel" of the critical state concept.

The critical state model prediction can be compared with contours inferred from laboratory data, which map out in the stress space the yield stresses corresponding to specific values of plastic volumetric strain ε_v^p (Fig. 7). To compare with the critical state model prediction, Baud et al. (2006) first fitted the initial yield stress data to an elliptic envelope, thus obtaining values of A_o, B_o and C_o. They then fitted the contours of subsequent yield stresses for selected values of $\Delta\varepsilon_v^p > 0$ to elliptic envelopes, imposing the equality $A = A_o$. These ellipses for Bentheim sandstone are also shown in Fig. 7, and the coordinates (C, B) so determined for the elliptic envelopes are plotted in Fig. 8c, which also include coordinate values that Baud et al. (2006) obtained for Adamswiller and Darley Dale sandstones using the same methodology. One can then test the "critical state" constraint by checking whether the coordinates (C, B) at the tip of the elliptic caps do indeed lie on a straight line. For those data that fall on a linear trend the critical state parameters m were evaluated by linear regression. It can be seen in Fig. 8c that the sandstone data fall on the predicted linear trends with "critical state" slopes of 0.54, 0.32 and 0.58, respectively. There seems to be a trend for the slope m to increase with decreasing porosity.

3.1.2. The DiMaggio-Sandler cap model

Evolution of the yield cap in DiMaggio and Sandler's (1971) model as a function of plastic volumetric strain is illustrated by the schematic diagram in Fig. 8b. In this scenario the parameters of the ellipse have more degrees of freedom since only one constraint is imposed here: the aspect ratio of the cap remains constant, so that

$$B/A = B_0/A_0 = R_0 \tag{5}$$

while the rock undergoes work hardening. To compare experimental data with this cap model, Baud et al. (2006) fitted the initial and subsequent yield stresses for selected values of $\varepsilon_v^p > 0$ to elliptic caps without imposing any constraints on either the width of the minor axis or aspect ratio. If indeed yield behavior of the sandstone was as predicted by the DiMaggio-Sandler cap model, then the value of R should remain the same as it strain hardens. The aspect ratios $R = B/A$ so determined for the yield caps of Bentheim sandstone are plotted in Fig. 8d.

Baud et al. (2006) observed that the yield stress data for Bentheim sandstone can be fitted reasonably well with elliptic caps up to plastic volumetric strain of 0.06. It can also be seen from Fig. 8d that in agreement with the DiMaggio–Sandler model, aspect ratios of the caps were approximately constant, with an arithmetic means of 1.16. They also concluded that the data for Berea sandstone are in basic agreement with the cap model, with a mean aspect ratio of 1.08. In contrast, the aspect ratios of Adamswiller and Darley Dale sandstones were inferred to decrease rapidly with increasing plastic strain, implying that the DiMaggio–Sandler cap model is not applicable.

3.2. Normality condition and associated flow rule

Other than the yield stresses a plasticity model should also specify how plastic strains evolve after the rock has yielded. In plasticity theory the conventional approach is to assume that the inelastic deformation follows Drucker's (1951) postulate of material stability, and accordingly a flow rule can be associated with the yield function by

$$\Delta\varepsilon_{ij}^p = \frac{\partial f}{\partial \sigma_{ij}}\Delta\lambda \tag{6}$$

where $\Delta\varepsilon_{ij}^p$ denotes the plastic increment of the strain tensor and $\Delta\lambda$ is a positive scalar. Since this equation implies that the plastic strain tensor will be oriented perpendicular to the yield surface, it is also called the "normality condition" (Desai and Siriwardane,

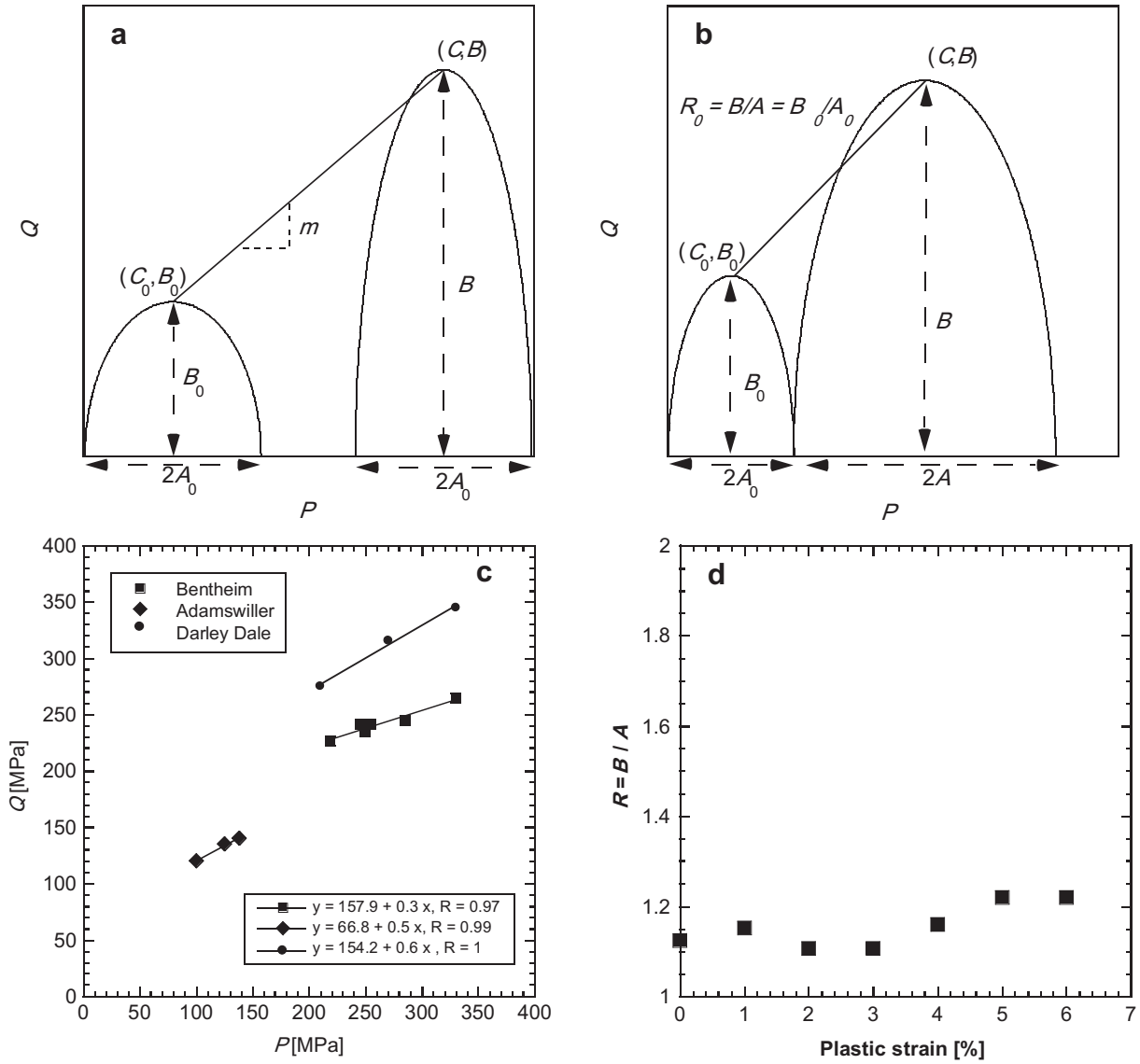


Fig. 8. (a) Schematic diagram illustrating evolution of the yield envelope in the effective mean stress – differential stress space for Carroll's (1991) critical state model. With increasing plastic volumetric strain, the peak of the yield surface with coordinates (C, B) traverses the critical state line of slope m. The width $2A_o$ of the elliptical yield surfaces remains constant. (b) Schematic diagram illustrating evolution of the yield envelope in the effective mean stress – differential stress space for DiMaggio and Sandler's (1971) cap model. The aspect ratio B/A of the elliptical caps remains constant with increasing plastic volumetric strain. (c) Coordinates for the peaks of the yield envelopes are shown by solid symbols. The critical state can be inferred from these experimental data: Linear fits to the solid symbols represent the critical state lines according to Carroll's (1991) model. The lines were obtained by linear regression. (d) Aspect ratio of the elliptical yield cap (shown in Fig. 7) as a function of plastic volumetric strain for Bentheim sandstone. According to the DiMaggio and Sandler's (1971) cap model the aspect ratio should remain constant.

1984; Davis and Selvadurai, 2002). While such an associated flow rule specifies all six components of the strain tensor, a conventional triaxial test provides independent measurements of only two strain components, primarily in terms of the axial strain and volumetric strain (or porosity change). The ratio between the volumetric and axial components of the plastic strain can be derived by substituting the yield function (2) into the normality condition (6). If we denote the volumetric strain increment by $\Delta\varepsilon_v^p$ and axial strain increment by $\Delta\varepsilon^p$, then their ratio is given by (Baud et al., 2006):

$$\frac{\Delta\varepsilon_v^p}{\Delta\varepsilon^p} = \frac{3}{(3A^2/B^2)(Q/(P-C))+1} \tag{7}$$

This implies that the volume contracts (or expands) for effective mean stress P greater (or less) than C. In other words a transition in deformation mode from compactancy to dilatancy occurs at the tip of the yield envelope located at (C, B). At this transition point the porous material is considered to be at the "critical state". However, it can be observed from Fig. 7 that some data points for compactive yield stress actually fall on the side of the ellipse $P < C$, thus contradicting the normality prediction that at relatively low mean stresses the sample would yield by dilatancy and not compactancy.

To further test the normality condition on experimental data, if we assume as before that the plastic component of volumetric strain is dominated by porosity change (so that $\Delta\varepsilon_v^p/\Delta\varepsilon^p \approx \Delta\Phi^p/\Delta\varepsilon^p$), then the model prediction (7) for $\Delta\varepsilon_v^p/\Delta\varepsilon^p$ can be compared with the "inelastic compaction factor" $\Delta\Phi^p/\Delta\varepsilon^p$ (Wong et al., 1997), that one can determine from laboratory data. Among four sandstones analyzed by Baud et al. (2006), compaction data of Darley Dale sandstone show best agreement with the prediction of associated flow rule. In Fig. 9a, the factor $\Delta\Phi^p/\Delta\varepsilon^p$ they determined

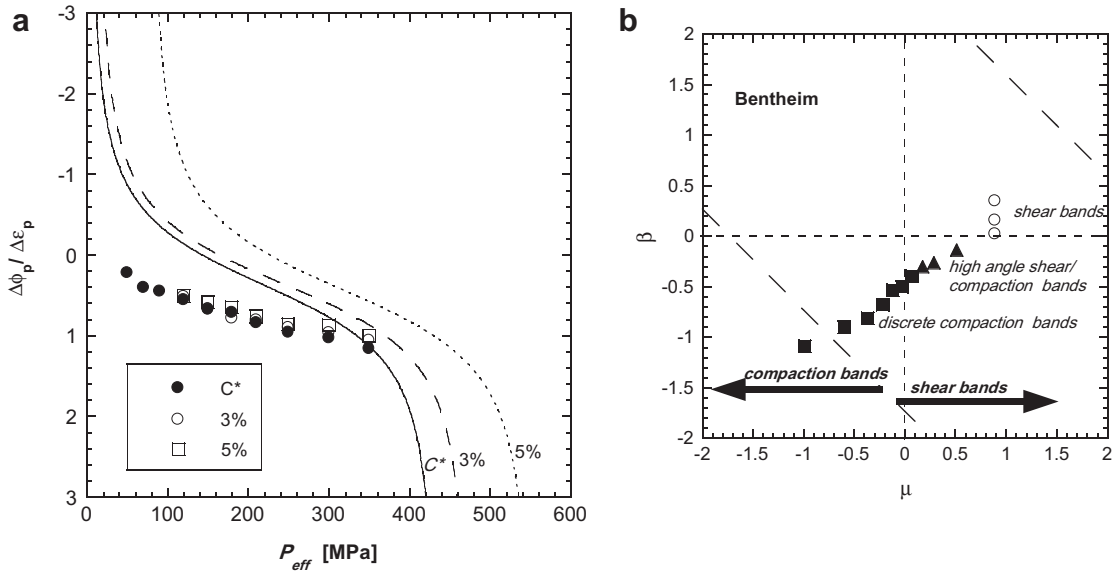


Fig. 9. (a) Comparison of experimental data with prediction of the normality condition in Bentheim sandstone. The inelastic compaction factor at the onset of shear-enhanced compaction C^* and increasing plastic volumetric strain as a function of the effective pressure at which a triaxial compression experiment was conducted. Solid lines represent the theoretical predictions for associative behavior according to Equation (7b). (b) Constitutive parameters for the non-associative model of Rudnicki and Rice (1975): (a) Bentheim sandstone. The solid circles, squares and triangles correspond to the dilatancy factor β as a function of internal friction coefficient μ at the onset of shear-enhanced compaction for samples that failed by the development of homogeneous cataclastic flow, discrete compaction bands and high-angle shear/diffuse compaction bands, respectively. The open circles correspond to the constitutive parameters at the peak stress for samples that failed by dilatant faulting. The three failure modes (compaction band, shear band and dilation band) predicted from bifurcation analysis of Issen and Rudnicki (2000) are separated by the diagonal lines $\mu + \beta = -3$ and $\mu + \beta = 3(2 - \nu)/(1 + \nu)$ where ν is Poisson's ratio (of elastic materials).

for Bentheim sandstone is plotted as a function of effective pressure for selected values of plastic volumetric strain, and for comparison the theoretical predictions according to (7) for the associated flow rules are shown as the continuous curves. The comparison shows that shear-enhanced compaction was considerably larger than that predicted for associative behavior, especially at relatively low mean stresses. This implies that the lateral expansion of a yielded sample was generally less than that prescribed by the normality condition. Similar discrepancy has been observed in relation to brittle failure of soil (Chen, 1984; Desai and Siriwardane, 1984) and compact crystalline rock (Senseny, 1986), for which an associated flow rule consistently overestimates the magnitude of dilatancy.

The two classes of constitutive models are widely used in geotechnical and geological applications since they involve a relatively small number of parameters. The normality condition is implicitly assumed in the original formulations of the critical state and cap models, although it has been relaxed in some implementations of the constitutive models. These constitutive models have been built into finite element codes such as ABAQUS or FLAC3D for the numerical simulation of inelastic compaction in porous rock (Vajdova and Wong, 2003; Grueschow, 2005; Sheldon et al., 2006), ceramic powder (Zeuch et al., 2001) and pharmaceutical powder (Wu et al., 2005). The study of Baud et al. (2006) underscores the necessity to first characterize experimentally the evolution of compactive yield behavior before selecting the appropriate constitutive model for a porous rock. The most systematic study has been on sandstones, although there are preliminary data on carbonate rocks (Elliott and Brown, 1985) that showed basic agreement with the normality condition, but more systematic studies of this nature are clearly warranted in the future.

3.3. Non-associated flow rule and localization analysis

The theoretical prediction of the inception of strain localization is sensitive to whether the normality condition is applicable. Given the experimental observations, in a localization analysis one should consider a non-associative model, which necessarily involves a plastic potential function and additional constitutive parameters. As an example we will focus on Rudnicki and Rice's (1975) model, which was originally formulated for analyzing the onset of shear localization in a dilatant frictional material. In this isotropic hardening model three constitutive parameters (in addition to two elastic moduli) were used for characterizing the inelastic and failure behavior: an internal friction parameter μ, a dilatancy factor β, and a hardening modulus h. The flow law is associative only if $\beta = \mu$.

As elaborated by Wong et al. (1997), the constitutive parameters can be extracted from triaxial compression experiment data in the following manner. If we assume that $\Delta\varepsilon_v^p \approx \Delta\Phi^p$, then the dilatancy factor β is related to the inelastic compaction factor by

$$\beta = -\sqrt{3}\,\frac{\Delta\Phi^p/\Delta\varepsilon^p}{\left(3 - \Delta\Phi^p/\Delta\varepsilon^p\right)} \tag{8a}$$

The parameter β can therefore be derived from the $\Delta\Phi^p/\Delta\varepsilon^p$ values evaluated from laboratory data as discussed above. The difference in sign convention between the dilatancy and compaction factors should be noted: if $\Delta\Phi^p/\Delta\varepsilon^p < 0$, then $\beta > 0$. As for the "friction" parameter μ it can be evaluated as $\sqrt{3}/3$ times the slope of the yield envelope in the P-Q space, or explicitly in terms of the parameters for an elliptic cap as

$$\mu = -\frac{B^2(P - C)}{\sqrt{3}A^2 Q} \tag{8b}$$

Using these equations Baud et al. (2006) evaluated these two parameters in the Rudnicki and Rice model from their initial yield data at the onset of shear-enhanced compaction. Their data for β and μ of Bentheim sandstone (with the failure mode indicated) are shown in Fig. 9b. For reference they also included data for samples that failed by dilatant faulting with development of shear bands.

The data show that as Bentheim sandstone undergoes the brittle-ductile transition, the failure mode evolves from shear band to compaction band to distributed cataclastic flow as the constitutive parameters β and μ decrease with increasing effective pressure. Bifurcation analyses (Rudnicki and Rice, 1975; Olsson, 1999; Issen and Rudnicki, 2000) specify the critical conditions for the onset of strain localization, thus providing a theoretical framework for understanding how this transition in failure mode arises from constitutive behavior. Fig. 9b summarizes the theoretical predictions for axisymmetric compression by mapping out the failure mode as a function of the constitutive parameters β and μ, according to Issen and Rudnicki (2000). If Poisson's ratio is denoted by ν, then shear bands are predicted to develop for

$$\sqrt{3} \le \mu + \beta \le \sqrt{3}(2-\nu)/(1+\nu) \tag{9a}$$

In contrast, dilation bands (subparallel to σ_1) would develop if

$$\mu + \beta > \sqrt{3}(2-\nu)/(1+\nu) \tag{9b}$$

and compaction bands would develop if

$$\mu + \beta < -\sqrt{3} \tag{9c}$$

Hence the localization analysis predicts that the brittle-ductile transition is associated with decreasing β and μ, in qualitative agreement with the laboratory observations. A similar conclusion applies to the data for several other sandstones (Wong et al., 2001; Baud et al., 2006). However, significant discrepancies exist between the quantitative predictions and laboratory data, in that β and μ data of many samples that failed by compaction banding fall within the theoretically inferred domain of shear banding (Fig. 9b).

This discrepancy is possibly due to the inadequacy of the constitutive model of Rudnicki and Rice (1975) to capture comprehensively the partitioning of multiple damage mechanisms, including the growth and coalesce of stress-induced microcracks, collapse of pores and crushing of grains. A more elaborate constitutive model will necessarily involve a number of additional parameters. Indeed systematic investigations in the laboratory have motivated the development of several such models that attempt to account for the possible influence of elastic–plastic coupling (Papamichos et al., 2000), two yield surfaces (Issen, 2000), anisotropy (Rudnicki, 2002) and grain breakage (Das et al., 2011). It was also postulated by Grueschow and Rudnicki (2005) that the yield cap for porous sandstone should evolve as a function of not only the plastic volumetric strain but also the shear strain. This feature was implemented in a numerical model formulated recently by Stefanov et al. (2011) to simulate the evolution of compaction bands.

4. Microstructural observations

Wong et al. (2004) and Paterson and Wong (2005) have reviewed the various techniques for characterization of the microstructure and damage in deformed rock samples. Given the diversity of preexisting microstructures induced by igneous, sedimentary and tectonic processes in porous rock, it is indeed surprising that their macroscopic mechanical behaviors in relation to the brittle-ductile transition are so similar, which raises the intriguing possibility that the operative micromechanical processes can be captured by a set of unified models applicable to porous rocks in general. However, systematic microstructural observations of deformed samples in the past few decades have demonstrated that different types of porous rock have fundamentally different processes of damage development, and accordingly distinct micromechanical models are required to capture these processes, even though they may predict qualitatively similar phenomenology of brittle-ductile transition.

4.1. Brittle faulting and damage evolution in porous rock

In the brittle faulting regime, extensive reviews of the microstructural evolution in laboratory deformed samples have been presented by Dresen and Guéguen (2004) and Paterson and Wong (2005), and in naturally deformed samples by Blenkinsop (2000) and Passchier and Trouw (2005). Hence we will only provide a brief synopsis, and the readers are referred to these references for a more detailed treatment.

Rocks are characterized by a large variety of microstructural attributes (grain size and shape; pore size and shape; spatial distribution of grains, pores, and cracks, etc.) As reviewed by Kemeny and Cook (1991), there are many mechanisms (such as preexisting sliding cracks, pores, elastic mismatch and dislocation pile-up) which can result in grain-scale concentration of tensile stress. Whether any one of these mechanisms can be activated in a particular rock depends on its mineralogy and pore geometry, as well as pressure and temperature conditions. In a porous rock, the pore space is typically made up of relatively equant pores and microcracks, both of which can be sites of tensile stress concentration and initiation of microcracking when an overall compressive stress is applied to the rock. In contrast, in a compact rock the pores seem to play a secondary role in initiating stress-induced cracking, possibly because both their numbers and sizes are smaller.

The onset of dilatancy is associated with the initiation of stress-induced cracking. Beyond the crack initiation stage, damage development in a compact and porous rock has some common features. As the stress-induced cracks propagate, they tend to align gradually with directions subparallel to the maximum compression direction. As the stress increases further, it induces more such wing cracks to initiate and propagate. Interaction and coalescence of these cracks ultimately lead to the onset and development of shear localization. The accumulation and localization of damage have been documented by microstructural observations, and they are also manifested by corresponding changes in physical properties, including sonic speeds, hydraulic permeability, electrical conductivity and acoustic emission (AE) activity (Dresen and Guéguen, 2004; Paterson and Wong, 2005).

4.2. Cataclastic flow, strain localization and damage evolution in porous rock

4.2.1. Siliciclastic rocks

Microstructural and AE observations have provided important insights into the spatial and temporal evolution of grain-scale damage during inelastic compaction of porous rock. In a sandstone, typically the AE activity shows an accelerated increase near the compactive yield stresses and peaked after the sample has undergone a significant amount of inelastic compaction (Zhang et al., 1990b; Wong et al., 1997; Baud et al., 2004; Fortin et al., 2006). AE activity in rock is a proxy for damage processes that include microcracking, frictional slip, and pore collapse. The evolution of damage associated with distributed cataclastic flow in porous sandstones has been characterized by Zhang et al. (1990b), Menéndez et al. (1996), Wu et al. (2000) and Mair et al. (2002). Intragranular cracking dominates the hydrostatic and triaxial compaction processes. The surge in AE activity at the onset of inelastic compaction marks the onset of "Hertzian fractures" that radiate from impinging grain contacts due to local tensile stress concentrations (Fig. 10a). These intragranular cracks extend across the grains, leading to grain crushing and ultimately pore collapse

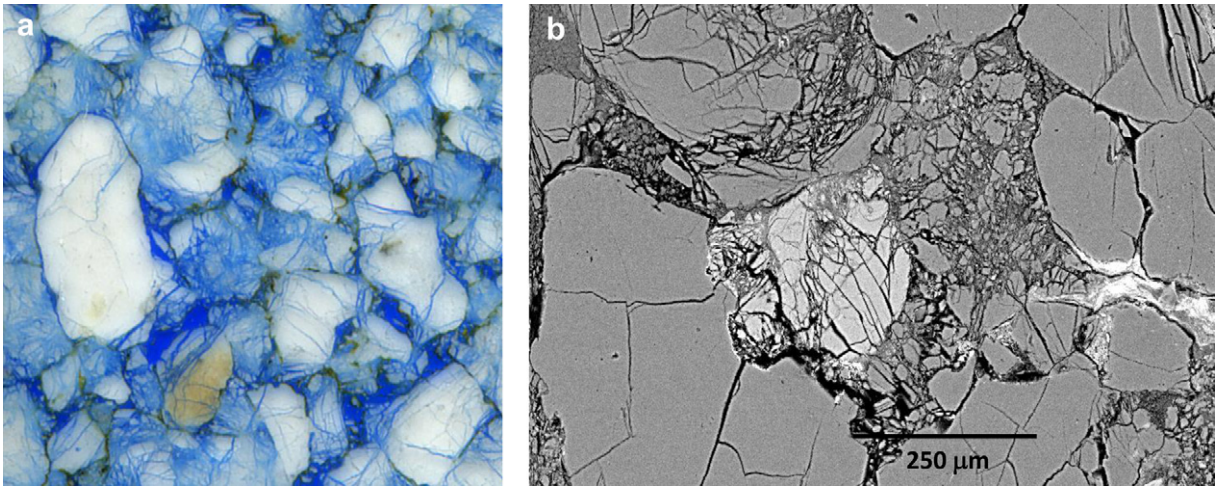


Fig. 10. Extensive grain crushing and pore collapse in samples of: (a) Boise (optical micrograph, width ~6.4 mm), and (b) Darley dale (SEM micrograph) sandstones failed by cataclastic flow.

(Fig. 10b). These micromechanical processes are manifested by corresponding changes in the sonic velocities (Fortin et al., 2007).

While earlier studies focused on the evolution of damage with distributed cataclastic flow, a number of recent studies have elucidated the micromechanical processes related to compaction localization in sandstone, using techniques including optical and scanning electron microscope observations (Baud et al., 2004; Vajdova et al., 2004b; Tembe et al., 2008), X-ray CT-imaging (Louis et al., 2006; Charalampidou et al., 2011) and AE locations (Fortin et al., 2006; Townend et al., 2008). To characterize the geometric attributes (such as thickness, tortuosity, and spacing) of discrete compaction bands, one must image a relatively large representative volume (which necessarily implies a relatively low resolution), and although X-ray CT can provide a nondestructive technique for imaging the 3-dimensional density distribution, it is difficult in this situation to resolve compaction localization solely from contrasts in the X-ray attenuation. To address this limitation, Louis et al. (2006) have demonstrated that cataclastic damage (due to grain crushing and pore collapse) would decrease both the skewness and dispersion of CT-values of an ensemble of voxels. Using the coefficient of variation for 27 voxels embedded in a cube with volume comparable to that of a grain, they were able to derive a local measure of damage which maps out the 3D geometric complexity of discrete compaction bands of the Diemelstadt sandstone (Fig. 11a).

Focusing on a small cylindrical volume (4 mm in diameter) of a Bentheim sandstone sample that failed at confining pressures of 300 MPa in nominally dry conditions (see Fig. 1d of Baud et al., 2004), we obtained an X-ray CT image with a refined resolution of 4 μm (Fig. 11b), which highlights the significant contrasts in porosity and grain size inside and outside a compaction band. A common microstructural observation is that the initiation and propagation of compaction bands seem to be controlled by grain-scale heterogeneities. The discrete bands made up of intensely comminuted grains typically extend laterally over ~2–3 grains, with thicknesses <1 mm (Fig. 11c).

4.2.2. Carbonate rocks and ignimbrite

Unlike the Hertzian fracture scenario in sandstone, inelastic compaction in limestone is associated with pore collapse that would initiate from stress concentrations around the surface of an equant pore, which induce a ring of localized damage in its periphery. We show here an example from a Tavel limestone sample that was hydrostatically compressed to beyond the critical pressure P^*. The macropore has a relatively circular cross-section with a diameter of ~20 μm, that is surrounded by a halo of cataclastic damage (Fig. 12a). A concentric rim of relatively intense damage has extended over a thickness of ~2 μm. While the spatial distribution of damage is circumferentially symmetric, its intensity decays with radial distance, such that the micritic matrix seems relatively undeformed beyond a thickness of ~6 μm. To gain a better perspective into the damage distribution around the pore surface, the sample was tilted slightly and the same area observed under the SEM in the secondary electron mode (Fig. 12b). The cavity is filled with what appear to be debris (from polishing the sample) mixed with epoxy, and the exposed pore surface is lined with a thin layer of microcracked materials.

Recent microstructural observations on micritic and allochemical limestones (Vajdova et al., 2010, 2012) indicate that the larger pores tend to collapse first, with cataclasis and microcracking as the dominant deformation mechanisms in the proximity of the collapsed pore. Unlike “plastic pore collapse” which initiates from crystal plasticity processes (Curran and Carroll, 1979), in a porous limestone intense microcracking would develop within a concentric ring surrounding the macropore, and comminuted fragments spall and fall into the void, in a scenario that Zhu et al. (2010) referred as “cataclastic pore collapse”.

Under conventional triaxial loading, the development of cataclastic damage would reflect the axisymmetric loading, as illustrated in Fig. 13a for a Tavel sample loaded to well beyond the compactive yield stress C^* at a confining pressure of 150 MPa (Vajdova et al., 2010). In contrast to the hydrostatically compacted sample, the spatial distribution of damage is asymmetric around the circumference, with the most intensive damage in two clusters that lie around the equator. Zooming into the damage zone, numerous cracks that had emanated from micropores and propagated sub-parallel to maximum compression direction can be observed. These pore-emanated cracks had begun to coalesce. For another macropore in this limestone sample, the comminuted fragments had spalled and fallen into the void (Fig. 13b).

Recently Vajdova et al. (2012) investigated the micromechanics of inelastic compaction in two allochemical limestones. Inelastic compaction in both limestones was associated with pore collapse that seems to initiate from stress concentrations at the surface of a pore, with cataclasis developed preferentially around the macropores, in qualitative agreement with observations on the micritic Tavel limestone. However, their observations also show that the

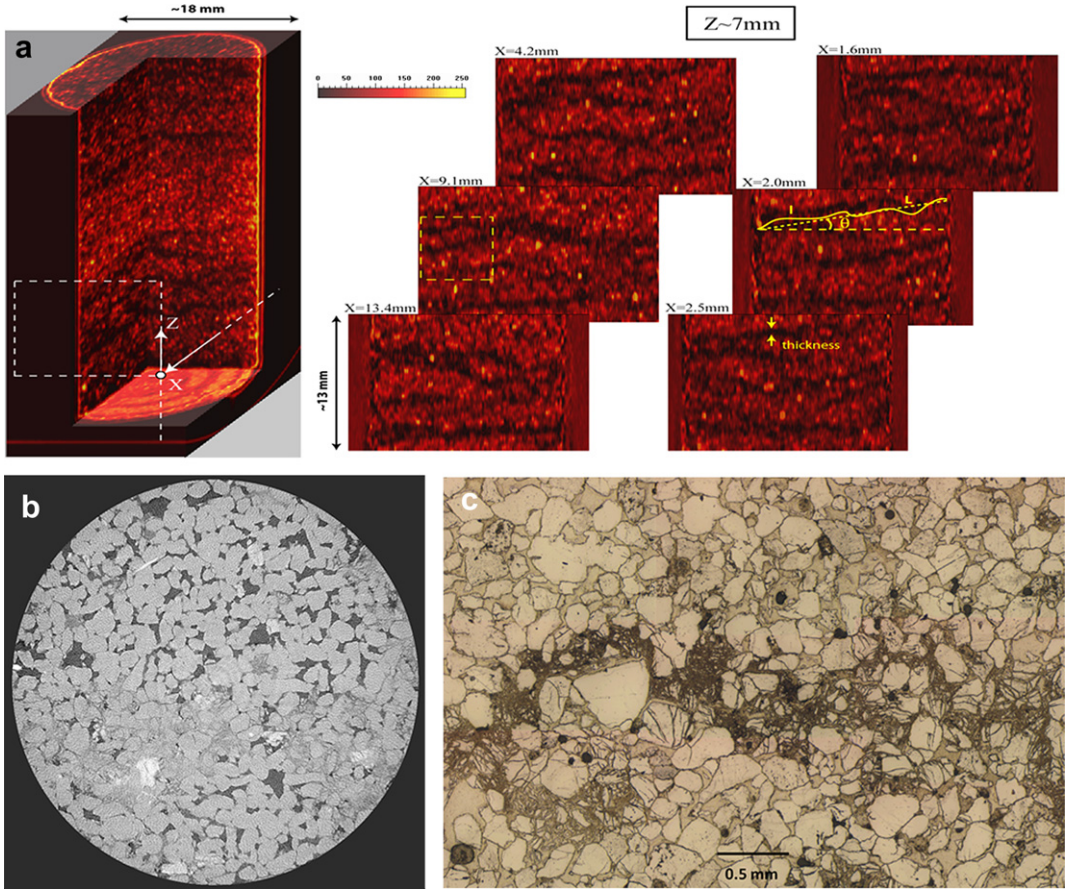


Fig. 11. (a) 3D block reconstruction in terms of the local coefficient of variation value of a sample of Dielmelstadt sandstone deformed at 130 MPa of effective pressure (Louis et al., 2006). A number of dark tortuous bands emanating from traces of light bands seen on the exterior surface of the sample. Six serial sections for the rectangular area marked in white. These images show the complex geometry of compaction bands in the third dimension. (b) CT data with resolution of 4 μm for a vertical slice passing through a compaction band in a sample of Bentheim sandstone deformed at a confining pressure of 300 MPa. A porosity reduction of 14.8% in the band was inferred from this image. (c) Mosaic of micrograph showing the details of a discrete compaction band. The width of the band is about two grains. Principal stress σ_1 was along the axial direction.

spatial distribution of damage in the allochemical limestones can be complicated by its uneven partitioning among the allochems, micrite and sparite. Fig. 14a and b show two examples of the damage in shear-compacted samples of Indiana limestone. Vajdova et al. (2012) concluded that many allochems remain relatively intact even after the cement has undergone significant microcracking, with the implication that significant strength contrast exists between the allochems and cement.

In their study of brittle-ductile transition in a porous ignimbrite, Zhu et al. (2011) observed a similar mechanism of cataclastic pore collapse for the development of inelastic compaction. Two sets of their data for a nonwelded tuff are presented here. Fig. 14c shows a ring of cataclastic damage extending over a radial distance of ~100 μm, that has developed around a macropore in a sample hydrostatically compacted beyond P^*. Fig. 14d shows the total collapse of a macropore in a sample that has undergone significant

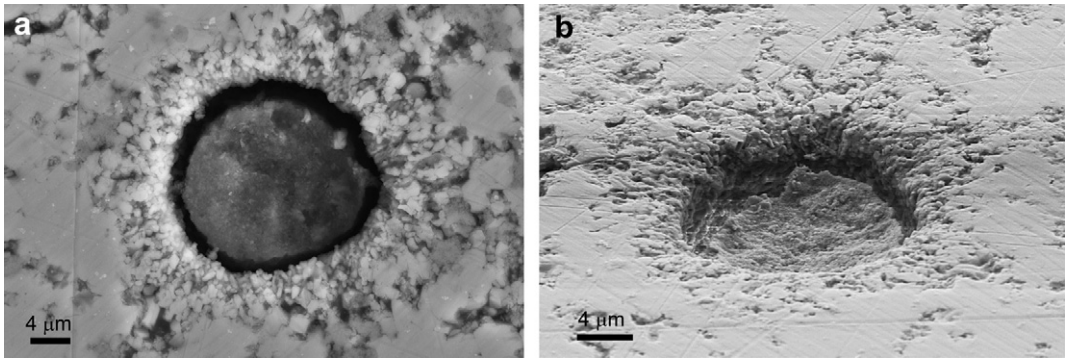


Fig. 12. (a) Backscattered SEM image of the hydrostatically compacted sample of Tavel limestone. A macropore surrounded by a halo of cataclastic damage was observed. The diameter of the macropore is ~20 μm, the concentric rim of particularly intense damage extended over a thickness of ~2 μm. (b) Secondary electron microscope image of the same macropore in (12a). The sample was rotated 90° anticlockwise and then tilted slightly. The cavity is filled with debris (from polishing the sample) mixed with epoxy, and the exposed pore surface is lined with a thin layer of microcracked materials.

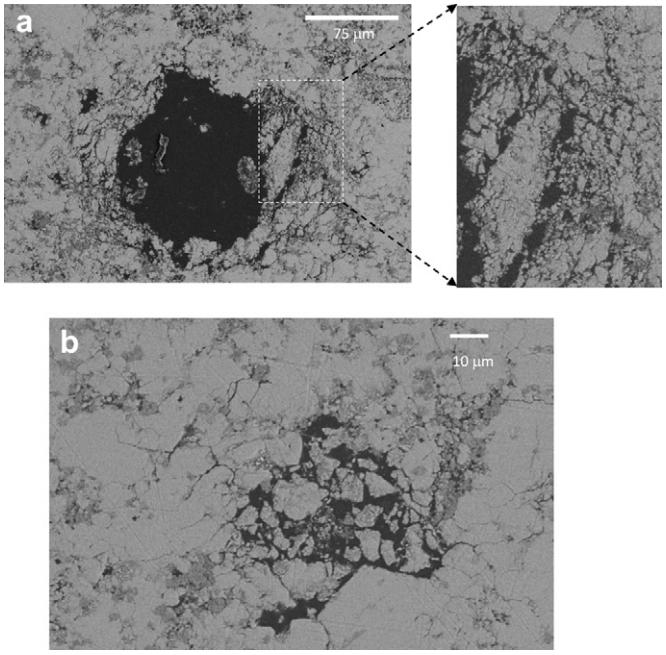


Fig. 13. Backscattered SEM images of samples of Tavel limestone failed by cataclastic flow (Vajdova et al., 2010). Direction of σ_1 is vertical. (a) This sample was deformed at 150 MPa of confining pressure and loaded to just beyond $C^{*\prime}$. One of the largest collapsed macropores visible in this sample. Numerous pore-emanated cracks propagated subparallel to σ_1. Intensive cataclastic damage around the equator was observed, with details inside the dashed rectangle shown in the zoomed-in figure on the right. (b) Collapse of spalled fragments into a macropore in the same sample. Stress-induced microcracks propagated to distances comparable to the pore dimension.

shear-enhanced compaction at an effective pressure of 30 MPa. The void space is now filled with crushed grains and spalled fragments.

Calcite requires relatively low shear stresses to initiate mechanical twinning and dislocation slip even at room temperature (Turner et al., 1954; Griggs et al., 1960). As noted by Vajdova et al. (2004a) the differential stress levels required for the development of inelastic compaction in most porous limestones are comparable to or exceed the shear stresses required to activate *e*-twinning in calcite. It should also be noted that twinning and microcracking are intertwined, in that the former can result in significant tensile stress concentration for the nucleation of microcracks (Olsson and Peng, 1976; Fredrich et al., 1989). Such an interplay of pore collapse, cleavage cracking and twinning is illustrated in Fig. 15a, for an Indiana limestone sample that had been hydrostatically compacted to 114 MPa beyond $P^* = 60$ MPa (Vajdova et al., 2012).

Under nonhydrostatic loading, the interaction between cataclasis and crystal plasticity can be quite complex, as illustrated in Fig. 15b for an Indiana limestone sample triaxially compressed at 20 MPa confining pressure (Vajdova et al., 2012). Twin lamellae emerging on a grain boundary may lead to microcracking in a neighboring grain. The twins evidently interacted with the cleavage or the cracks, although it is difficult to categorically conclude whether the twinning induced cleavage crack initiation or the opposite. In the relatively compact Tavel limestone, Vajdova et al. (2010) documented a progressive increase of twinning density with increasing strain in the cataclastic flow regime. In their study of the use of calcite twinning for paleotress estimation, Rowe and Rutter (1990) considered three different measures of twinning activity and investigated comprehensively their dependences on grain size, stress, strain, strain rate and temperature. One of their measures is “twinning incidence”,

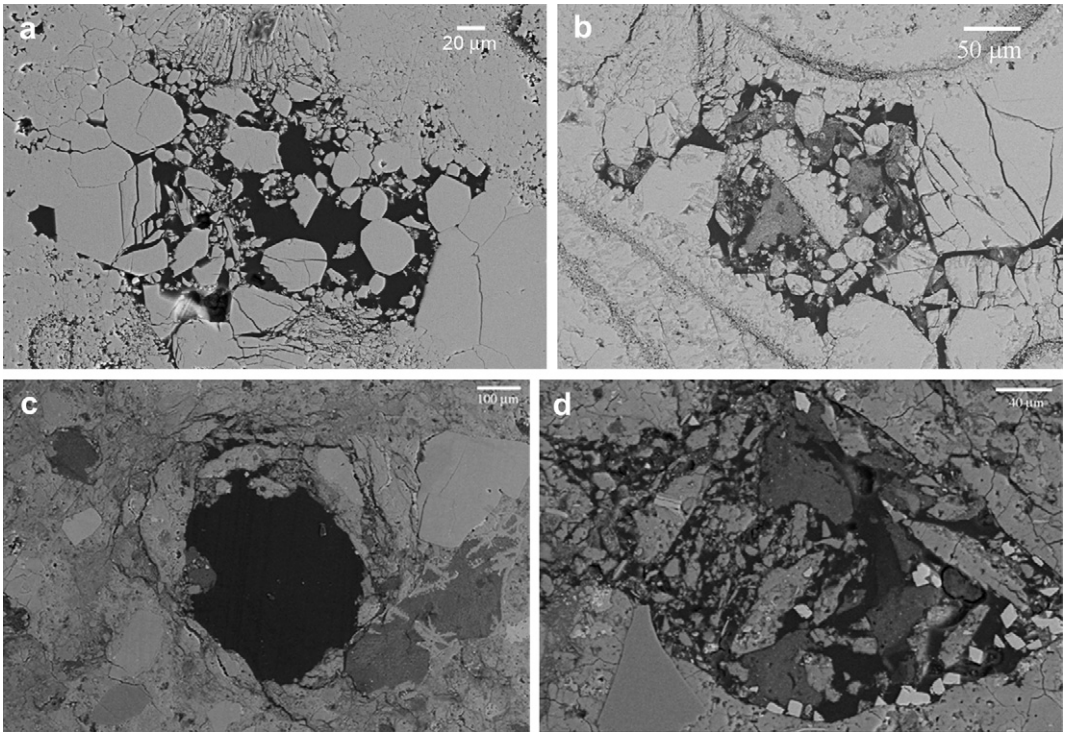


Fig. 14. Backscattered SEM images of samples of Indiana limestone (Vajdova et al., 2012) and Alban Hills tuff (Zhu et al., 2011). Direction of σ_1 is vertical. (a) Collapsed macropore within a cemented region that is surrounded by five allochems in a sample of Indiana limestone deformed at a confining pressure of 20 MPa and loaded beyond the onset of shear enhanced compaction C^*. (b). Pervasive collapse of macropores accompanied by cataclastic damage in their peripheries in a sample of Indiana limestone deformed at a confining pressure of 20 MPa and that failed by cataclastic flow. (c) Sample of Alban Hills tuff deformed at 20 MPa of effective pressure and stressed to beyond the compactive yield stress C^*. A macropore in matrix with a diameter ~250 μm was observed to collapse. A thin layer of crushed grains in the periphery of the macropore was formed by the propagation and coalescence of stress-induced microcracks. (d). Collapse of spalled fragments into a macropore in a sample of Alban Hills tuff deformed at 30 MPa of effective pressure and stressed to beyond the compactive yield stress C^*. Crushed grains had fallen into the interior of the macropore.

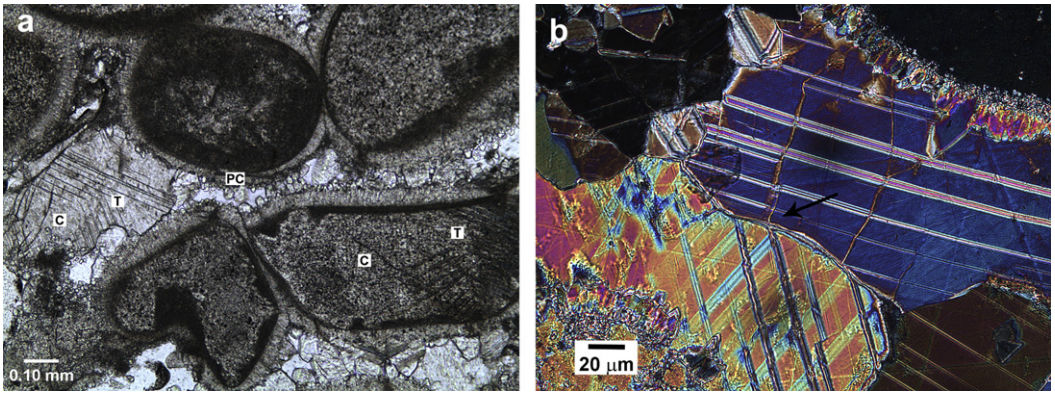


Fig. 15. Interplay between microcracking and mechanical twinning. Optical micrographs, direction of σ_1 is vertical. (a) Indiana limestone sample deformed hydrostatically to 114 MPa beyond $P^* = 60$ MPa, in plain light. Pore collapse (PC), cleavage cracks (C) and twinning (T) were observed. (b) Sample of Indiana limestone deformed triaxially at confining pressure 20 MPa, crossed nicols. Emergence of twin lamellae on a crystal surface leads apparently to microcrack formation.

defined to be the percentage of grains in a given grains size class interval which contain optically visible twin lamellae. They observed that this measure of twinning activity increases with grain size (at constant level of differential stress) and linearly with stress (at a given grain size). Furthermore all three measures of twinning activity were observed to be independent of strain, strain rate and temperature. Since Rowe and Rutter (1990) focused on compact carbonate rocks, it requires further systematic investigation to assess the extent to which their conclusions on calcite twinning activity can be generalized to limestones of significantly higher porosity.

5. Micromechanical modeling of brittle faulting and inelastic compaction

A micromechanical model should capture the key aspects of damage evolution as inferred from microstructural and AE observations. Taking into account the microstructure and based on continuum mechanics concepts (such as linear elastic fracture mechanics and plasticity theory), a micromechanical analysis is formulated with the ultimate objective of arriving at yield and constitutive relations that describe the macroscopic inelastic and failure behaviors. Motivated by the microstructural observations summarized in the previous section, a number of micromechanical models have been developed to describe various mechanisms that have been identified with the brittle-ductile transition in porous rock. We will first review those models that have been proposed for brittle faulting, and then the cataclastic flow regime.

Two end-members are often invoked to describe the microstructure of a porous rock for such micromechanical analysis. In the first scenario, the rock is assumed to be strongly cohesive so that the intergranular boundaries play little role and the microstructure can be idealized in terms of an inclusion model with microcracks and equant pores embedded in an elastic continuum. The preexisting cracks and pores may act as stress raisers for the initiation of extensile cracks which can then propagate through the porous medium, and the mechanical behavior is controlled by the evolution of such damage. The initiation and propagation of these stress-induced cracks can be analyzed using linear elastic fracture mechanics. Under high confinement, the pores may also undergo compactive yield leading to pore collapse, in which case the damage development can be analyzed using plasticity theory. In the other scenario, the rock is assumed to be weakly cohesive, such that the bonds at the grain contacts are readily ruptured by applied stress, allowing neighboring grains to slip and rotate relative to each other. In this granular model the representative element volume consists of a group of grains that can move relative to each other, exchange neighbors, and undergo dilation, comminution or other processes involved in accommodating the relative motions.

5.1. The brittle faulting regime

Micromechanical modeling of brittle faulting (for both compact and porous rocks) has been reviewed by Kemeny and Cook (1991), Dresen and Guéguen (2004), and Paterson and Wong (2005). Here we will mostly highlight recent developments that are pertinent to our topic.

5.1.1. Pore-emanated and sliding wing crack models

For the first scenario of an inclusion model, the representative element volume is idealized as a pore or crack embedded in an elastic continuum (Fig. 16a). If we first consider the case of a relatively equant pore, the applied macroscopic stress can induce localized stress concentration at the pore and nucleate cracks which emanate at the pore surface and propagate parallel to the maximum compression direction. Two-dimensional models based on axial cracks emanating from cylindrical pores have been considered by Sammis and Ashby (1986), Isida and Nemat-Nasser (1987), and Wimmer and Karr (1996). The stable propagation of pore-emanated cracks initially results in dilatancy and strain hardening. As the cracks propagate further, they begin to interact and if their mean spacing is sufficiently small, crack propagation becomes unstable, possibly leading to crack coalescence in a scenario analogous to microstructural observations reviewed above.

Recent applications to porous carbonate rocks and tuffs have shown that this pore-emanated cracking model can provide a relatively simple micromechanical interpretation of the dependence of the uniaxial compressive strength (UCS) on porosity and pore size. With reference to Sammis and Ashby's (1986) model, Zhu et al. (2010) derived an analytic approximation for this dependence. The damage mechanics model considers an elastic medium pervaded by circular holes of uniform radius r. As the applied stress increases, a point is reached when the stress intensity factor of a small crack on the circular surface attains the critical value K_{IC}, at which point extensile cracks would initiate from the poles and propagate to a certain distance parallel to the σ_1 direction. As the stress-induced cracks propagate to longer distances with increasing stress, they interact with one another to induce an additional

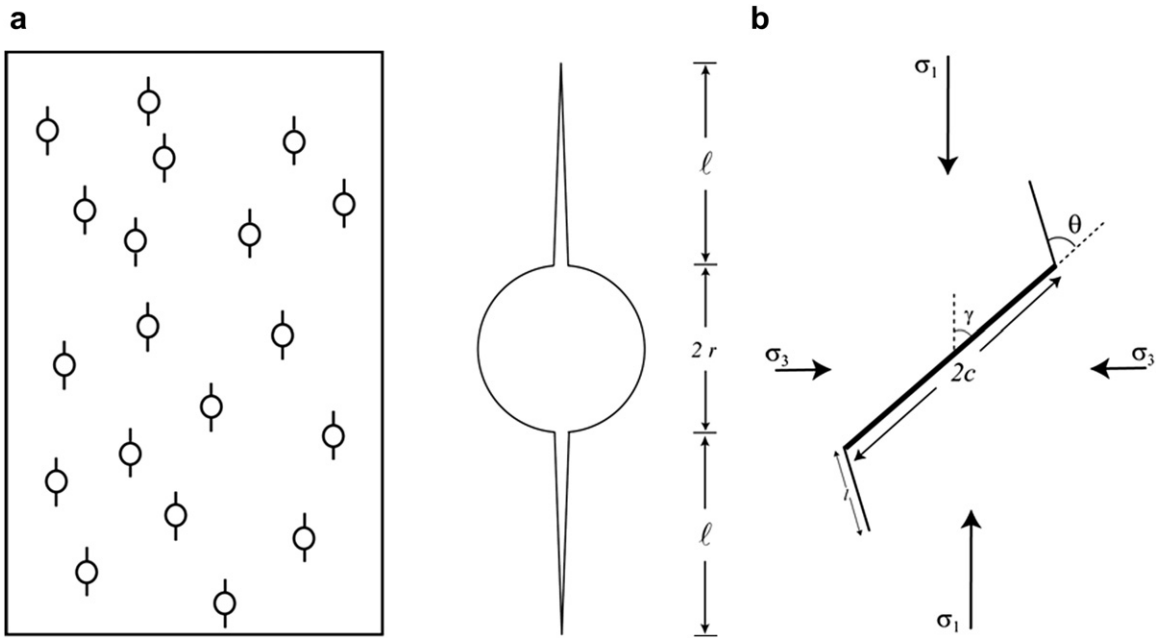


Fig. 16. (a) Sammis and Ashby's (1986) pore-emanated cracking model. A two-dimensional elastic medium is permeated by circular holes of uniform radius *r*. Cracks emanate from circular pores when subjected to remotely applied axial stress. The stress-induced cracks have propagated to a length *l*. Propagation and coalescence of pore-emanated cracks lead to instability and macroscopic failure. (b) Sliding wing crack model of Ashby and Sammis (1990) for the development of dilatancy and instability in the brittle faulting regime. Directions of the maximum and minimum principal stresses are indicated. The sliding and wing cracks are at angles of γ and θ with respect to the σ_1 direction, respectively.

tensile stress intensity, ultimately leading to an instability with coalescence of the pore-emanated cracks at the peak stress level (which corresponds to the UCS for uniaxial loading). Zhu et al. (2010) obtained an analytic estimate of the UCS σ_u (valid for Φ ranging up to 40%):

$$\sigma_u = \frac{1.325}{\Phi^{0.414}} \frac{K_{IC}}{\sqrt{\pi r}} \tag{10}$$

According to this pore-emanated cracking model, the UCS decreases with increasing porosity following a power law, and it scales with the fracture toughness and inverse square root of the pore radius. Zhu et al. (2010) compared the analytic prediction (10) with UCS data of micritic and allochemical limestones (made up of 80% or more calcite) with porosities ranging up to 30%. They concluded that the limestone data fall into two groups. Data for compact micritic limestones (with porosity <15%) can be bracketed by the two theoretical curves for $K_{IC}/\pi r = 35$ and 75 MPa, whereas data for allochemical and porous micritic limestones can be bracketed by the two theoretical curves for $K_{IC}/\pi r = 5$ and 22 MPa (Fig. 17a). If one assumes a K_{IC} value of 0.2 MPa $m^{1/2}$ based on fracture mechanics measurements on calcite (Atkinson and Meredith, 1987), then the pore-emanated cracking model together with the unconfined strength data imply that the compact micritic limestones have average pore size of $r = 2$–10 μm and the allochemical and porous micritic limestones have an average pore size of $r = 26$–500 μm, which Zhu et al. (2010) considered to be in basic agreement with their microstructural observations.

Zhu et al. (2011) recently compiled UCS data for nonwelded tuff and made a similar comparison with the micromechanical model (Fig. 17b). Except for some extremely weak tuffs at the high-porosity end, they observed that most of the laboratory data can be bracketed by the two theoretical curves corresponding to $K_{IC}/\pi r = 5$ and 35 MPa. Zhu et al. (2011) suggested using a value of $K_{IC} = 0.3$ MPa $m^{1/2}$. Then according to the analytic approximation the laboratory data for UCS of nonwelded tuffs are bracketed by average pore size ranging from $r = 23$ μm to 1.14 mm, which they considered to be in basic agreement with the microstructural observations.

We also compile here in Fig. 17c UCS data for sandstone, which are from the extensive tabulations of Kwasniewski (1989) and Jizba (1991). We have lumped together the medium- to coarse-grained sandstone data of Kwasniewski (1989) as one group. Jizba's (1991) samples are all fine-grained, and together with Kwasniewski's (1989) fine-grained data they are shown as open circles. The sandstones cover a range of feldspar and clay contents, and they are bracketed by the two theoretical curves corresponding to $K_{IC}/\pi r = 10$ and 50 MPa. If we follow Zhang et al. (1990a) to use a value of $K_{IC} = 0.3$ MPa $m^{1/2}$ for sandstone grains, then according to the analytic approximation the laboratory data for sandstones are bracketed by average pore size ranging from $r = 11.4$ μm to 0.29 mm.

More general results for biaxial loading were also derived by Sammis and Ashby (1986), but their model predicts that the confinement can readily inhibit brittle failure at a very low lateral stress level, with the implication that the pore-emanated cracking mechanism is of limited relevance to compressive failure under confinement (Wong, 1990). This limitation does not apply to the sliding wing crack model (Horii and Nemat-Nasser, 1986; Ashby and Sammis, 1990; Kemeny and Cook, 1991), which has also been adopted to analyze brittle failure behavior in porous sandstone, limestone, and tuff. The model considers sources of tensile stress concentration that are located at the tips of preexisting cracks (Fig. 16b). The applied far-field stresses induce a shear traction on the crack plane (of length 2*c*), and if the resolved shear traction exceeds the frictional resistance along the closed crack, frictional slip occurs which also induces tensile stress concentrations at the two tips that may nucleate and propagate wing cracks parallel to the σ_1 direction. As wing cracks propagate to longer distances, they interact and ultimately coalesce to result in an instability. The model predicts that the maximum and minimum principal stresses

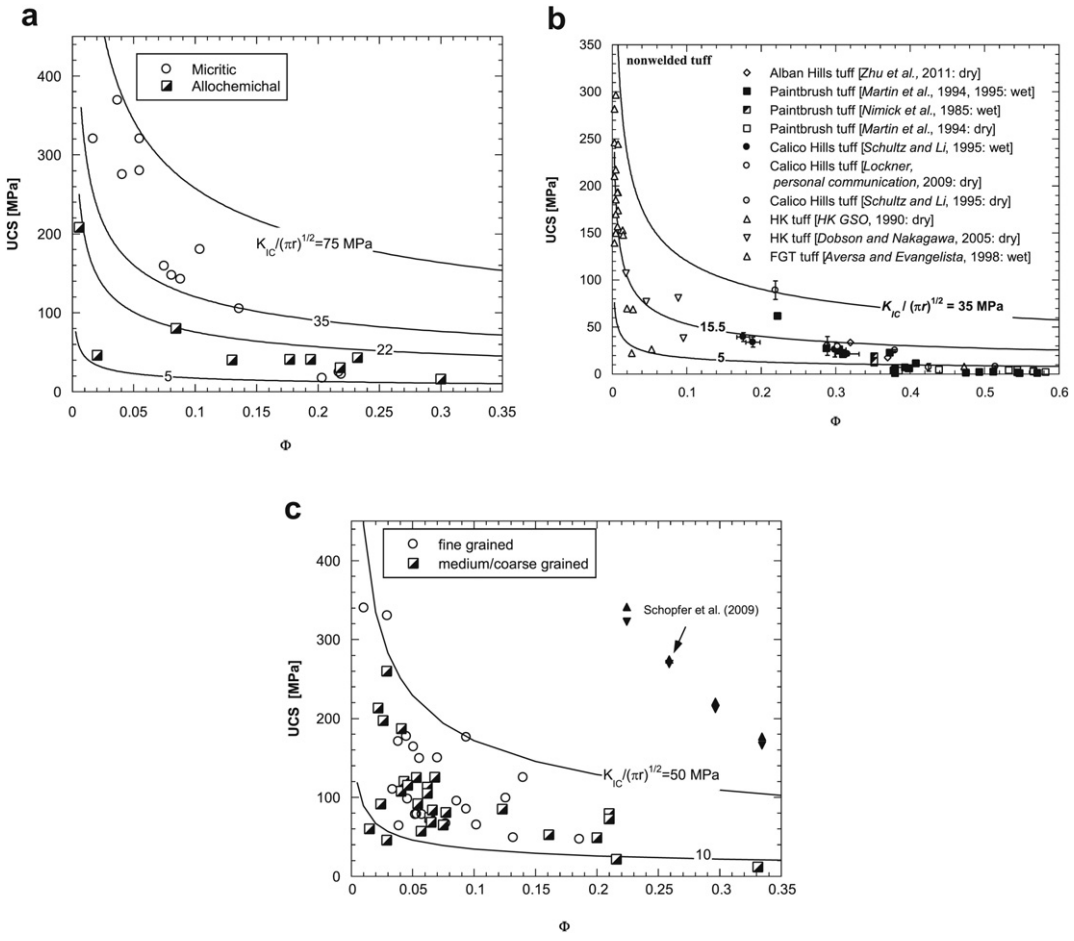


Fig. 17. Comparison of theoretical predictions with laboratory data on unconfined compressive strength (UCS) of (a) micritic (circles) and allochemical (square) limestones, (b) nonwelded tuffs, and (c) fine grained (circles) and medium/coarse grained (squares) sandstones. Theoretical curves of UCS as a function of porosity for different values of $K_{IC}/\pi r$ are plotted. Results from DEM simulation (Schöpfer et al., 2009) are also presented in Fig. 17c.

at the onset of dilatancy and peak both fall on linear trends, and therefore if mechanical data on these critical stresses (C′ and Coulomb failure parameters) are available, then the model predictions can be tested and relevant micromechanical parameters inferred.

At a fixed lateral stress, the sliding wind crack cracking model predicts that the peak differential stress scales with the parameter $K_{IC}/\pi c$ (Horii and Nemat-Nasser, 1986; Ashby and Sammis, 1990; Kemeny and Cook, 1991). The brittle strength is also predicted to decrease with increasing "initial damage" (or "crack density") which is characterized by a nondimensional parameter D_o, that is proportional to the number of preexisting cracks per unit area and the crack length squared (in the two-dimensional model). For a broad range of rock types (including granite, sandstone, limestone, gabbro, gneiss and basalt), values of μ inferred from dilatancy onset and peak stress data are in the range of 0.3–0.7, and of D_o in the range of 0.1–0.4 (Ashby and Sammis, 1990; Gupta and Bergström, 1998; Myer et al., 1992; Pestman and van Munster, 1996; Baud et al., 2000a, b; Vajdova et al., 2004a, 2010, 2012; Zhu et al., 2011).

With reference to the sliding wing crack model of Ashby and Sammis (1990), Zhu (2010) recently developed analytic approximations for the Coulomb failure parameters (cohesion, UCS, internal friction coefficient) in terms of the microstructural parameters (μ, $K_{IC}/\pi c$ and D_o), which allows one to compare laboratory data with the theoretical predictions without tedious numerical simulations. In particular, the UCS and internal friction coefficient μ_i (corresponding to an internal friction angle φ) are given by:

$$\sigma_u = \frac{1.346}{\sqrt{1+\mu^2}-\mu}\frac{K_{IC}}{\sqrt{\pi c}}D_0^{-0.256} \tag{11a}$$

$$\mu_i = \tan\varphi = -0.062 - 0.501\log D_o + (0.895 + 0.142D_o)\mu; \tag{11b}$$

The UCS, which scales with the normalized fracture toughness $K_{IC}/\pi c$, is sensitively dependent on the initial damage D_o and friction coefficient μ of the sliding crack. The internal friction coefficient is predicted to be linearly related to the coefficient μ, and the difference between the two coefficients has an approximately logarithmic dependence on the square root of the initial damage D_o.

5.1.2. Granular models

When the bonding at grain contacts is ruptured and neighboring grains are enabled to slip and rotate relative to each other, a micromechanical model is needed which accounts for this particulate motion since it plays a dominant role in the development of dilatancy or compaction. The representative element volume now consists of a group of grains that can move relative to each other, exchange neighbors, and undergo dilation, comminution or other processes involved in accommodating the relative motions.

The elastic and inelastic behavior of a particulate system has been studied extensively in soil mechanics using analytic and numerical models. Analytic models of spheres in elastic contact have provided useful insight into the mechanical behavior (Johnson, 1985). In recent years, the discrete element method (DEM) has proved to be an effective numerical tool for simulating the micromechanics of failure in unconsolidated materials, such as soil, sediment and fault gouge (e.g., Cundall and Strack, 1979; Antonellini and Pollard, 1995; Hazzard and Mair, 2003). In a DEM model local deformation on the grain scale can arise from elastic strains at impinging grain contacts or relative slip among grains, which collectively contribute to a global strain field in the granular assembly that may be homogeneously distributed or localized. The granular material is represented as an assemblage of spherical particles, although other geometrical shapes (ellipsoids, blocks) have also been considered in a few cases.

The DEM has also been extended to a bonded granular material (Potyondy et al., 1996; Potyondy and Cundall, 2004), with contacts that can be bonded and thus unable to slip. Typically a contact is assigned a tensile and shear strength, and if either of these stress thresholds is attained the bond would rupture, thus opening the possibility for relative slip between neighboring grains when the local tangential force is sufficiently high to overcome the frictional resistance at the ruptured grain contact. Such an approach has been adopted to model the micromechanics of failure in both compact and porous rocks (Hazzard et al., 2000). With appropriate calibration of the input parameters, the DE model can simulate the progressive development of strain hardening/softening and shear localization as observed in the laboratory.

Schöpfer et al. (2009) have recently conducted a systematic DEM study of the influence of porosity and initial damage on the brittle strength of a porous rock. In their 3-dimensional DEM models, a series of (conventional) triaxial compression and extension tests were performed on assemblages of spherical particles, with different size distributions and porosities. They observed an approximately linear decrease of the brittle strength with increasing porosity, which is almost independent of the particle size distribution. This seems to be in discrepancy with the nonlinear relation (10) for the pore-emanated cracking model. However, since they could consider only a narrow range of porosity ($\sim$22–38%) limited by the spherical particle packing, their DE simulations do not necessarily imply that the linear trend would persist over a broader range of porosity. For comparison we include in Fig. 17c the UCS results from Fig. 10a of Schöpfer et al. (2009). It can be seen that the DE model predictions would roughly correspond to the prediction of Equation (10) for $K_{IC}/\pi r \sim 100$ MPa, which is comparable to relatively high (tensile and shear) bond strengths of 100 MPa adopted by Schöpfer et al. (2009). It is possible that the DEM simulations would be in better agreement with the laboratory data in this porosity range if a significantly lower bond strength (by an order of magnitude) were used. Schöpfer et al. (2009) also considered different proportions of uncemented particle contacts (as a proxy for pre-existing cracks). The influence of initial damage (crack density) on the Coulomb failure parameters inferred from the DE simulations is qualitatively similar to prediction of the sliding wing crack model discussed earlier.

5.2. The cataclastic flow regime

Microstructural observations have shown that the micromechanics of cataclastic flow is quite different in clastic and carbonate rocks. In a clastic rock such as sandstone, inelastic compaction in a laboratory sample derives primarily from grain crushing initiated by the stress concentrations at grain contacts, which induce intragranular cracks to radiate in a conical pattern towards the interior of the impinging grains (Fig. 10). Therefore it is more appropriate to use a granular model for this micromechanical process. In contrast, observations in limestones have shown that inelastic compaction is associated with pore collapse, that seems to initiate from stress concentrations at the surface of an equant pore, which induce a ring of localized damage in its periphery (Fig. 12). For this scenario, it is more appropriate to consider an inclusion model.

5.2.1. Plastic and cataclastic pore collapse models: hydrostatic compaction

An inclusion model that has been developed and widely used for porous material compaction is the plastic pore collapse model (Bhatt et al., 1975; Gurson, 1977; Curran and Carroll, 1979), which considers the collapse of spherical pores induced by crystal plasticity processes such as dislocation slip and deformation twinning operative in the proximity of the pore surface. In this scenario, a concentric shell of plastic deformation would develop and transform the pore geometry from spherical to ellipsoidal with an overall decrease in volume. For the relatively compact Solnhofen limestone, Baud et al. (2000a) observed that its yield cap can be interpreted using Curran and Carroll's (1979) plastic pore collapse model. However, Vajdova et al. (2004a) concluded that the agreement of this plastic pore collapse model with the compactive yield behaviors of more porous limestones is quite limited. As summarized earlier, microstructural observations indicate at least two major discrepancies. First, pore collapse seems to first initiate at the larger pores, while the theoretical model predicts that it is equally likely for collapse to initiate from a large or a small pore. Second, cataclasis and microcracking (rather than crystal plasticity) seem to be the dominant deformation mechanisms in the proximity of a pore that has collapsed. Relatively intense cracking would develop with a concentric halo surrounding the pore, and comminuted fragments may spall and fall into the void.

Zhu et al. (2010) referred to this process as "cataclastic pore collapse" and developed a micromechanical model for its initiation. Since microstructural observations (Vajdova et al., 2010, 2012) showed that plasticity (twinning and dislocation slip) become important beyond a certain level of plastic strain, this model in only applicable to the initial yielding in limestone. Similar to previous pore collapse models (Bhatt et al., 1975; Gurson, 1977; Curran and Carroll, 1979), the pores are idealized as spherical in shape. An externally applied stress field induces local stress concentration at the surface of a pore, and yielding initiates when the local stresses satisfy a specified failure criterion. Whereas the von Mises (or Tresca) yield criterion is typically adopted in a plastic pore collapse model to describe failure by crystal plasticity mechanisms, brittle failure in the cataclastic pore collapse model obeys the Coulomb or Drucker–Prager criterion.

The mechanics of compaction is analyzed with reference to a representative element volume made up of a macropore embedded in an effective (porous) medium. If we first consider hydrostatic loading, the principal stresses $S_1 = S_2 = S_3 = P_c$ acts remotely on the external boundary of the element volume. It can be shown that local stresses in the vicinity of the pore are such that yielding will first occur at the spherical surface (Fig. 18a). With reference to a cylindrical coordinate system (ρ, θ, z), the local stresses there are given by:

$$\sigma_{\rho\rho} = 0 \tag{12a}$$

$$\sigma_{\theta\theta} = \sigma_{zz} = 3P_c/2 \tag{12b}$$

This stress state corresponds to an "unconfined compression", with the minimum principal stress (along the radial direction) equal to zero. If a microcrack were to initiate, it is expected to

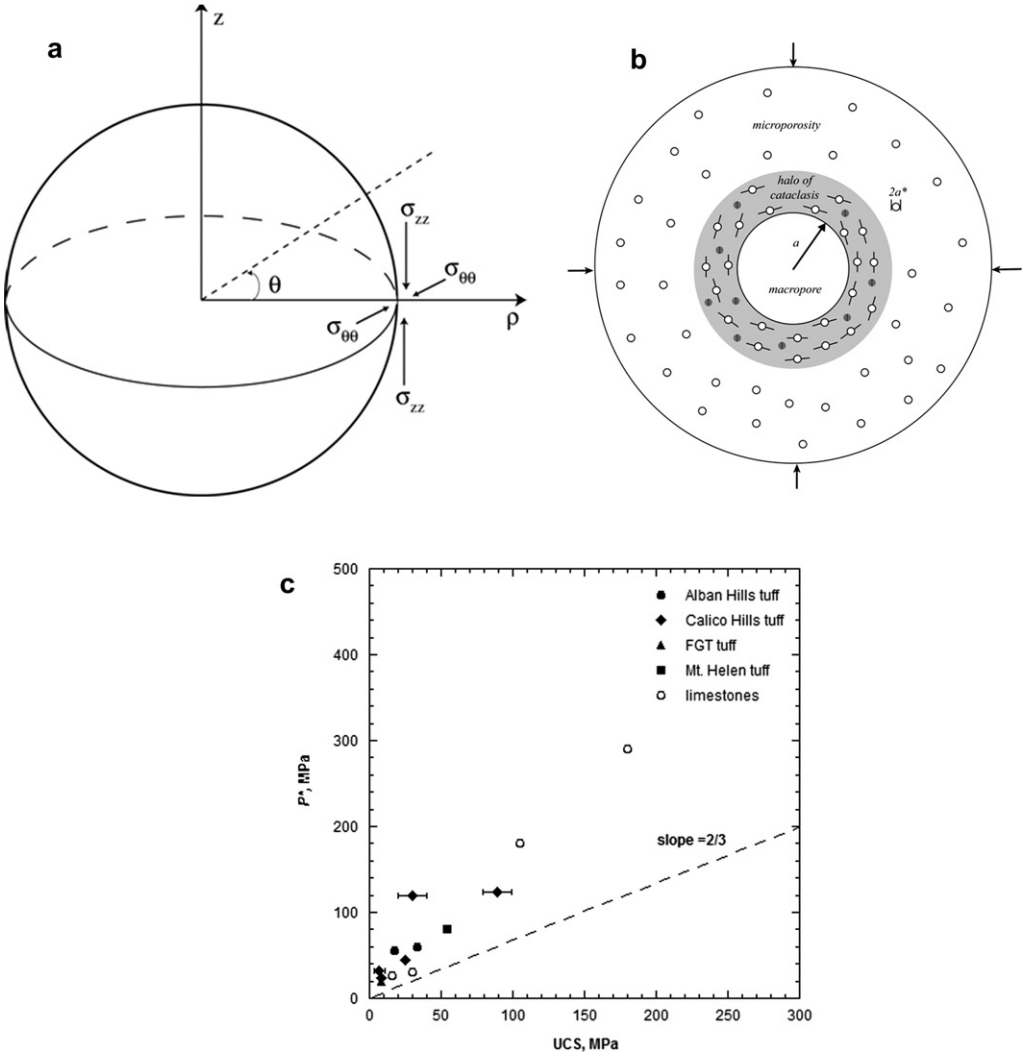


Fig. 18. Cataclastic pore collapse model of Zhu et al. (2010). (a) Local stress field at the vicinity of the macropore. The local principal stresses σ_{zz} and $\sigma_{\theta\theta}$ act along the axial and azimuthal directions, respectively. Because of the boundary conditions in the pore surface, the radial stress $\sigma_{\rho\rho}$ is zero. (b) Schematic diagram of a representative volume element of radius b. A macropore of radius a is surrounded by an effective medium made up of many micropores of radius a^*. Remote principal stresses are represented by S_1, S_2, and S_3. Stress-induced pore-emanated cracking in the vicinity of the macropore will lead to its collapse. (c) The critical pore collapse pressure P^* is plotted versus the uniaxial compressive strength UCS on nonwelded tuff samples of Zhu et al. (2011) (close symbols) and limestone samples of Zhu et al. (2010) (open symbols). The dashed line corresponds to a slope of P^* versus UCS equal to 2/3. The standard deviations of data on Calico Hills tuff are shown as error bars.

propagate along a surface that is perpendicular to the minimum principal stress (i.e., transverse to the radial direction), as indicated in Fig. 18b. If one adopts the Mohr-Coulomb failure criterion (which is independent of the intermediate principal stress), the failure stress for the equal biaxial stress configuration (12) is identical to that for uniaxial compression. Accordingly, initial yielding (which corresponds to the onset of cataclastic pore collapse) will occur when the maximum principal stress is equal to the UCS σ_u^* of the effective medium:

$$\sigma_{\theta\theta} = \sigma_{zz} = \sigma_u^* \Rightarrow P_c = P^* = 2\sigma_u^*/3 \tag{13}$$

A similar analysis can be performed for the Drucker–Prager criterion, which turns out to give an identical result for a remotely applied hydrostatic loading (Bhatt et al., 1975; Curran and Carroll, 1979; Zhu et al., 2010).

In previous analyses (Bhatt et al., 1975; Curran and Carroll, 1979), it is implicitly assumed that the effective medium has a UCS (and other failure parameters) identical to those of the bulk sample, which would require $\sigma_u^* = \sigma_u$. If indeed this assumption is valid, then (13) would imply that a plot of the pore collapse pressure P^* versus the UCS σ_u of the bulk sample falls on a linear trend with a slope of 2/3. To test the validity of this assumption, Zhu et al. (2011) compiled laboratory data for the critical pore collapse pressure P^* and UCS σ_u of porous limestones and tuffs (Fig. 18c). It can be seen that the data all plot above the dashed line (with slope 2/3), which implies that $P^* = 2\sigma_u^*/3 > 2\sigma_u/3$, and therefore $\sigma_u^* > \sigma_u$. In other words, the effective medium is inferred to have a UCS that is much higher than that of the bulk sample. The parameter σ_u^* cannot be measured in the laboratory, and one has to estimate its value from an appropriate micromechanical model.

To resolve this discrepancy between the UCS of the effective medium and bulk rock, Zhu et al. (2010) appealed to the dual porosity concept, which has been accepted widely in carbonate sedimentology (Choquette and Pray, 1970; Pittman, 1971; Anselmetti et al., 1998; Baechle et al., 2008). In light of the microstructural observations in limestones (Figs. 12 and 13) that cataclastic failure within the damage halo surrounding the macropore develops in a scenario analogous to the pore-emanated cracking model (with a UCS given by Equation (10)), we would expect the

effective medium either is less porous than the bulk rock, or has smaller pores.

The total porosity Φ is assumed to be the sum of macroporosity Φ_M and microporosity Φ_m, made up of large and small pores with average diameters a and a^*, respectively. The effective medium is modeled as a porous medium that contains a population of micropores (Fig. 18b). Since the macropores have been excluded from it, the effective medium has smaller pores and a porosity less than the total porosity; accordingly its UCS σ_u^* is greater that the UCS σ_u of the bulk sample which includes the macropores. The average micropore size a^* is assumed to be relatively small in comparison to average macropore size a and linear dimension b of the representative element volume, such that $b >> a >> a^*$. Brittle failure in the effective medium is assumed to be described by Sammis and Ashby's (1986) pore-emanated cracking model, and therefore its UCS σ_u^* is given by Equation (10) with $r = a^*$ and, on substituting into Equation (13), we obtain the following relation between the pore collapse pressure P^* and total porosity:

$$P^* = \frac{2}{3}\sigma_u^* = \frac{0.883}{\Phi_*^{0.414}}\frac{K_{IC}}{\sqrt{\pi a^*}} = \frac{0.883}{\Phi^{0.414}}S^* \tag{14a}$$

with

$$S^* = \frac{K_{IC}}{(\Phi_*/\Phi)^{0.414}\sqrt{\pi a^*}} \tag{14b}$$

where Φ_* denotes the porosity of the effective medium, which is related to the macroporosity and microporosity by $\Phi_* = \Phi_m/(1-\Phi_M) \approx \Phi_m$.

Hence this micromechanical model for the initiation of cataclastic pore collapse in a dual porosity medium predicts that while there is an overall decrease of the critical pressure with increasing porosity, the compactive yield stress is also dependent on the parameter S^* which characterizes the cooperative effect of micropore size, fracture toughness and partitioning of microporosity and macroporosity. Zhu et al. (2010) compiled laboratory data for limestones (made up of 90% or more calcite) with porosities ranging up to 30% (Fig. 19). To help define the overall trend, they also included data for chalk (with porosities ranging from 38% to 45%) that were compiled by Vajdova et al. (2004a). The data fall between two limiting curves accordingly to (14). The two samples with lowest porosities (Solnhofen and Tavel limestones) lie on the upper curve corresponding to $S^* = 130$ MPa. Since $\pi a^* = (K_{IC}/S^*)^2(\Phi/\Phi^*)^{0.828}$, the micropore diameter for this upper limit can be estimated to be $a^* \geq (K_{IC}/S^*)^2/\pi = 0.6$ μm, assuming a K_{IC} value of 0.2 MPa $m^{1/2}$. If $S^* = 130$ MPa, the predicted P^* for porosity down to 3% is 490 MPa, which implies that the local stresses at the spherical pore are $\sigma_1 = \sigma_2 = 3P^*/2 = 735$ MPa. While microcracks may nucleate from the submicron pores under such a high stress, there is also a possibility that crystal plasticity processes become dominant. The P^* data for samples with porosities beyond ~23% fall on a plateau given by the lower limit $S^* = 20$ MPa, which implies that $a^* \geq 32$ μm, in basic agreement with microstructural observations (Zhu et al., 2010; Vajdova et al., 2012).

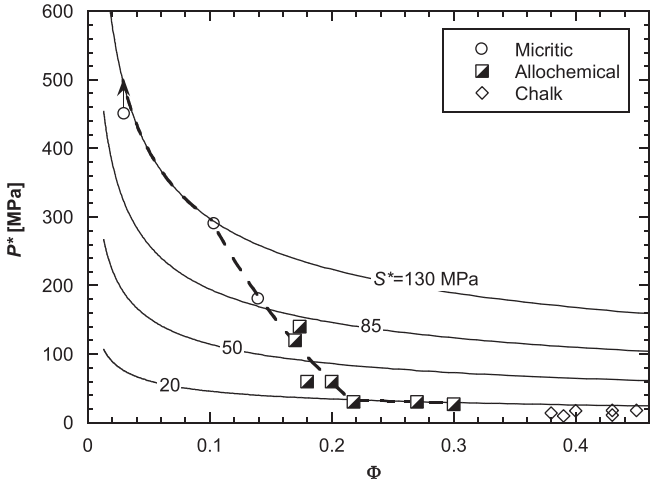


Fig. 19. Comparison of theoretical predictions with laboratory data on critical pressures for pore collapse (P^*) of limestones. Theoretical curves of P^* as a function of porosity for four different values of S^* (10) are plotted. Experimental data of micritic, allochemical limestones, and chalk are shown as circles, squares, and diamonds, respectively. They are bounded by upper and lower limits of $S^* = 130$ MPa and $S^* = 20$ MPa. The dashed curve with three segments corresponds to the empirical equation for S^* as a function of porosity proposed by Zhu et al. (2010).

Zhu et al. (2011) subsequently conducted an analogous interpretation of the compaction data for porous tuff, obtaining similar agreement between the model and laboratory data. In addition to the pore-emanated cracks, they also analyzed the effect of sliding wing cracks on the UCS of the effective medium, using the analytic approximation (11). Their analysis of the two mechanisms indicated that either the pore-emanated cracking or sliding wing crack model is consistent with their mechanical data and microstructural observations on Alban Hills tuff. The assumption was made that the two mechanisms are decoupled, but in reality they probably operate as coupled processes that result in cataclastic damage leading to pore collapse.

5.2.2. Cataclastic pore collapse model: triaxial compression

Two complications arise when we go beyond hydrostatic loading. The local stress field in the vicinity of a pore is more complicated under nonhydrostatic loading. Furthermore, predictions of critical yielding stresses are fundamentally different according to whether Coulomb or Drucker–Prager criterion is adopted. Consider a remote stress field $S_1 > S_2 = S_3$ applied to the representative volume element (Fig. 18a), corresponding to a conventional triaxial compression test in the laboratory. With reference to a cylindrical coordinate system (ρ, θ, z), the local stress distribution is such that initial yielding will occur first along the equator of the sphere (at $z = 0$ and $\rho = a$), where the principal stresses are given by (Timoshenko and Goodier, 1951).

$$\sigma_1 = \sigma_{zz} = \frac{3}{2}\left[\frac{(9-5\nu)}{(7-5\nu)}(S_1 - S_3) + S_3\right], \tag{15a}$$

$$\sigma_2 = \sigma_{\theta\theta} = \frac{3}{2}\left[\frac{(5\nu-1)}{(7-5\nu)}(S_1 - S_3) + S_3\right], \tag{15b}$$

$$\sigma_3 = \sigma_{\rho\rho} = 0 \tag{15c}$$

Substituting the (local maximum and minimum) principal stresses (15a) and (15c) into the Coulomb criterion, Zhu et al. (2010) derived the critical stress state C^* at the onset of shear-enhanced compaction:

$$S_1 - S_3 = \frac{(7-5\nu)}{(9-5\nu)}\left(\frac{2}{3}\sigma_u^* - S_3\right) = \frac{(7-5\nu)}{(9-5\nu)}(P^* - S_3) \tag{16a}$$

where ν denotes Poisson's ratio of the effective medium. If we define $P = (S_1 + 2S_3)/3$ and $Q = S_1 - S_3$, then the stress state C^* can also be expressed as

$$Q = \frac{3(7-5\nu)}{10(2-\nu)}(P^* - P) \tag{16b}$$

This predicts that a plot of Q versus P falls on a straight line with a slope that falls on a narrow range of 0.9–1.05 for $\nu = 0$–0.5. However, most laboratory observations do not support this amazingly simple prediction of a linear yield envelope, since most porous rock data indicate yield caps with curvature, except for an isolated study of Bleurswiller sandstone (Fortin et al., 2006) which shows an apparently linear cap.

Noting that the apparent discrepancy may be due to the neglect of the intermediate principal stress in the Coulomb criterion, Zhu et al. (2010) next considered the Drucker–Prager criterion which involves all three principal stresses. For conventional triaxial compression, they derived this quadratic relation between the remotely applied mean stress P and differential stress Q at the onset of shear-enhanced compaction:

$$\left(\alpha-\beta^2\gamma^2\right)Q^2+\left[\left(3\beta-4\beta\gamma^2\right)P-2\beta\gamma(1-2\gamma)P^*\right]Q + (1-2\gamma)(P-P^*)[(1+2\gamma)P+(1-2\gamma)P^*] = 0 \quad (17)$$

with $\alpha = 100(7-13\nu+7\nu^2)/[9(7-5\nu)^2]$, $\beta = 100(1+\nu)/[9(7-5\nu)]$, and $\gamma = 2\sin\phi/3 + \sin\phi$, where ϕ denotes the angle of internal friction used in the Coulomb criterion.

For comparison with data on porous limestones and tuffs, Fig. 6c shows the theoretical predictions of (17) according to the Drucker–Prager yield criterion (for a Poisson's ratio of 0.2 and three different angles of internal friction). Zhu et al. (2010) have shown that overall the effect of Poisson's ratio on the cap is relatively small. The nonlinear caps based on Drucker–Prager criterion are in qualitative agreement with laboratory data. The cap is predicted to expand with decreasing friction angle ϕ, and the highest differential stresses are associated with the cap for $\phi = 0$ (corresponding to the von Mises criterion), which shows the best quantitative agreement with experimental data, albeit at somewhat lower levels. For this limiting case, the differential stress Q as a function of the mean stress P is given by:

$$\frac{Q}{P^*}=\frac{3}{20}\frac{(7-5\nu)}{(7-13\nu+7\nu^2)}\left[\sqrt{4(7-13\nu+7\nu^2)-27\left(\frac{(1-\nu)P}{P^*}\right)^2}-(1+\nu)\left(\frac{P}{P^*}\right)\right] \quad (18)$$

with the pore collapse pressure P^* as a function of porosity given by (14) or (6) for an effective medium dominated by micropores. The experimental data agree better with the theoretical prediction for very low values of ϕ. The analysis suggests that brittle failure in such an effective medium with numerous micropores and fine microcracks has a pressure sensitivity significantly lower than that of the bulk sample. To be consistent with the experimentally determined yield caps, the model would require the effective medium to fail as a cohesive and pressure insensitive material.

5.2.3. Hertzian fracture and discrete element models

Both AE activity and microstructural studies on deformed sandstone samples consistently show that the critical pressure P^* is a manifestation of the onset of grain crushing and widespread microcracking within the granular structure. Therefore one would expect that the critical pressure is related to microstructural parameters and that one should use fracture mechanics in a granular model to tackle this problem.

Noting that the magnitude of P^* is primarily controlled by the porosity Φ and grain size R, Zhang et al. (1990a) formulated a Hertzian fracture model, in which the porous rock is idealized as a randomly packed assemblage of spherical particles of several distinct sizes. Under hydrostatic loading, the neighboring grains are in normal contact (Fig. 20) and tensile stress concentration is induced in the vicinity of the circle of contact between two impinging grains (Johnson, 1985). The maximum tensile stress is attained at the perimeter of the contact area, and the stress intensity factor K_I of a preexisting flaw located at the contact region is approximated by that of an edge crack. Hertzian fracture can initiate at the critical pressure P^* if K_I at a local contact reaches a critical value given by the fracture toughness K_{IC}. This micromechanical model for grain crushing predicts that

$$P^* = \frac{2.2}{E^2}\frac{(1-\nu^2)^2}{(1-2\nu)^3}\frac{K_{IC}^3}{(\alpha\Phi R)^{3/2}} \quad (19)$$

where E is the Young's modulus, ν is the Poisson's ratio, and α is the ratio of the initial crack length to grain radius.

If one makes the plausible assumption that the preexisting flaw dimension scales as the grain dimension R, then this initiation condition (19) would imply that the critical pressure P^* scales with the porosity and grain radius in accordance with $P^* \propto (\Phi R)^{-3/2}$. Wong et al. (1997) presented a compilation of the grain crushing pressure P^* as a function of initial porosity Φ and grain radius R, for porous siliciclastic rocks as well as unconsolidated materials (including sand and glass spheres). The mechanical data are in reasonable agreement with the theoretical prediction (19), with values of α on the order of 10^{-4}.

It should be noted that this Hertzian fracture model has limitations. First, it does not explicitly include the influence of other microstructural parameters such as cementation and clay content. Second, it is essentially an effective-medium type of model that neglects the heterogeneity of elastic contact force and statistics of fracture (Brzesowsky et al., 2011). Third, the model considers only the initiation of microcracks at grain contact. It does not address the question of whether the subsequent crack propagation is stable (Lawn, 1998), nor does it account for the mechanical interaction among grain contacts. Notwithstanding these limitations, this micromechanical model for the onset of grain crushing has proved to be a useful first-order description of inelastic compaction in

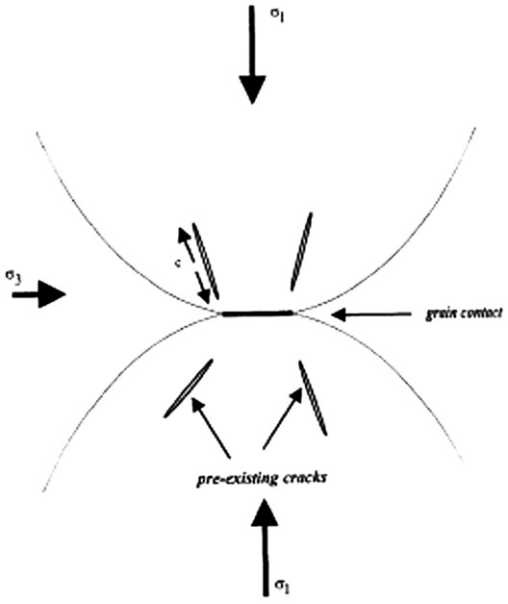


Fig. 20. Schematic diagram illustrating the Hertzian fracture model of Zhang et al. (1990a) for the onset of grain of crushing and cataclastic flow.

clastic rock. Since one end of the compactive yield cap is "anchored" at a location of the P axis corresponding to P^*, this implies that if a rock requires a relatively high effective pressure for grain crushing to occur, then the critical stress states for the onset of shear-enhanced compaction are also expected to be high. The compactive yield stresses C^* are expected to depend on porosity and average grain size in a manner similar to (19).

Beyond the onset of grain crushing, the development of cataclastic flow and possibly compaction localization involve complex interactions in a granular system. In recent years, there have been studies using the discrete element method for simulation of compaction band development (Li and Holt, 2002; Wang et al., 2008; Marketos and Bolton, 2009). The simulations of Wang et al. (2008) indicate that the development of discrete compaction bands is promoted in a relatively homogeneous granular aggregate (Fig. 21), while diffuse band growth and distributed cataclastic flow are preferred modes of compaction in a more heterogeneous system. Their results and results from network simulations (Katsman et al., 2005) agree with the suggestion of Klein et al. (2001) that discrete compaction band is predominant in the Bentheim sandstone possibly because of its relatively homogeneous mineralogy and well-sorted grain sizes.

6. Discussion

Conventionally the brittle-ductile transition is analyzed using an empirical criterion based on the pressure sensitivity of compressive strength. A transition to ductile flow is predicted to occur when the strength as a function of pressure (or mean stress) has a slope that deviates from the relatively steep slope in the brittle faulting regime (Mogi, 1966; Byerlee, 1968). The implementation of such criteria can be ambiguous, since the "strength" in the ductile regime evolves with strain hardening and is not well defined, and accordingly one has to arbitrarily assign it to be the stress attained at a fixed percentage of strain. In this review, we have focused on researches that go beyond this empirical approach, with the objective of arriving at firmer mechanical bases for fundamental understanding of the brittle-ductile transition in porous rock. Experimental data on the phenomenology of brittle-ductile transition, its constitutive modeling, microstructural characterization of the evolution of damage, as well as the micromechanical modeling of these phenomena have collectively provided useful insights into the physics and chemistry of the brittle-ductile transition. On one hand, conventional concepts from plasticity theory in conjunction with bifurcation analysis provide first-order macroscopic description of the inelastic and failure behavior and the intriguing phenomenon of compaction bands. On the other hand, damage mechanics and discrete element modeling provide useful tools for analyzing the micromechanics.

Compressive failure of porous sandstone has been investigated in the laboratory for a century following the seminal study of von Karman (1911). In parallel, extensive structural geology studies have been conducted to investigate the rich diversity of deformation and strain localization structures found in sandstone formations. From the field perspective a number of recent reviews (e.g., Borja and Aydin, 2004; Aydin et al., 2006; Fossen et al., 2007; Schultz and Fossen, 2008) have elaborated on the many attributes of failure mode and deformation microstructure which have been mapped in the field and are analogous to those observed in laboratory deformed samples. A similar conclusion seems also to apply to the limited field studies that have been conducted in tuff formations (Evans and Bradbury, 2004; Riley et al., 2010). Hence, systematic investigations of porous rock deformation under controlled conditions in the laboratory have provided useful insights into the geomechanical and tectonic processes.

The investigation of compaction band in sandstone is an example of how synergy can be derived from complementary researches in structural geology, experimental rock deformation and theoretical mechanics. In many respects, the geometric attributes of discrete compaction bands as observed in the laboratory (and reviewed above) are morphologically similar to compaction bands that have been documented in the field. Notwithstanding these similarities, Sternlof et al. (2005) emphasized that there are at least two apparent discrepancies between the failure modes and structures observed in the laboratory and in the field. First, the stress level for compaction band formation (on the order of 10 MPa) that they inferred in the Valley of Fire, Nevada is lower than that required in the laboratory (on the order of 100 MPa) for the development of discrete compaction bands. Second, the dimensions (thickness and length) as well as the damage intensity they measured in the field seem appreciably lower than those in the laboratory deformed samples. Aspects of these discrepancies were explained by Tembe et al. (2008) using a fracture mechanics model for compaction band propagation.

Synthesizing the field and laboratory data, they observed that the thickness and length of compaction bands obey a quadratic

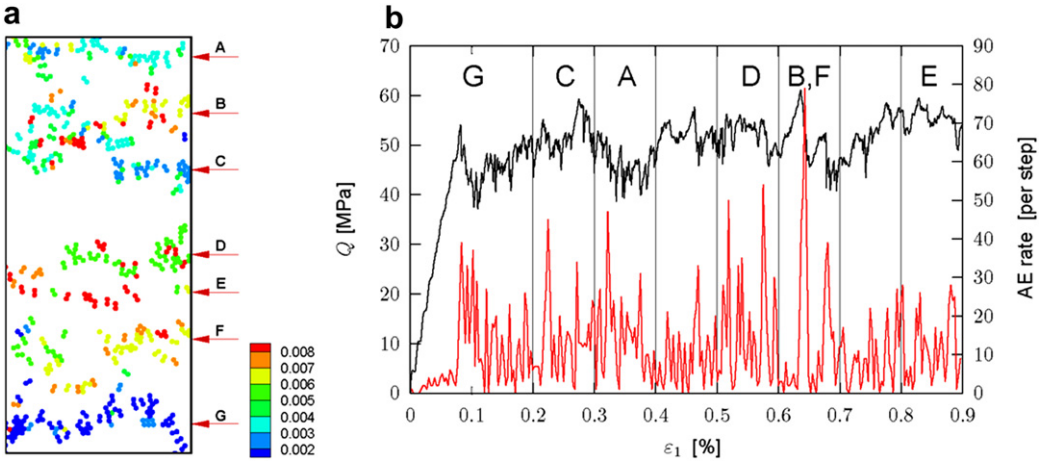


Fig. 21. Results of the DEM modeling of Wang et al. (2008) for the sequential development of discrete compaction bands in a well sorted assembly of bonded disks. AT least seven bands can be identified as distinct tabular clusters in: (a) the spatial distribution of damage, which can be identified with stress drops and AE surges in: (b), showing the differential stress (black line) and AE rate (rate line) as functions of axial strain. Color codes in the figure are in units of axial strain.

relation, where the thickness scales approximately with the square root of the band length. The mechanical basis for such a scaling relation was provided by Rudnicki (2007), who investigated the fracture mechanics of the propagation of a combined "anti-crack/anti-dislocation". Tembe et al. (2008) demonstrated that if Rudnicki's (2007) model applies to the propagation of compaction bands, then their thickness and length would indeed obey the quadratic relation, and furthermore the stress level involved is predicted to be inversely proportional to compaction band thickness. As illustrated in Fig. 22, the laboratory and field data on stress and compaction band thickness are consistent with this prediction of an "anti-crack/dislocation" model, with values of critical strain energy release rate in the range of 2–80 kJ/m^2, comparable to laboratory estimates of the nominal compaction energy of 6–43 kJ/m^2 inferred from measurements on circumferentially notched samples (Vajdova and Wong, 2003; Tembe et al., 2006). Schultz (2009) recently reported field measurements of length and thickness of compaction bands in Navajo Sandstone from Utah that are in basic agreement with this interpretation.

In comparison to clastic rocks, the field observations of such processes in carbonate rocks are limited. It seems that the failure mode is not as diversified, and the microstructure is such that cataclastic processes are more difficult to characterize in carbonate rocks. In an allochemical limestone, the damage development can be complicated by its uneven partitioning among the allochems, micrite and sparite. Vajdova et al. (2012) observed that in Indiana limestone, many allochems remain relatively intact even after the cement has undergone significant microcracking and fragmentation, with the implication that significant strength contrast exists between the allochem and cement. This observation is in agreement with some of the more systematic field and laboratory studies of carbonate compaction which have recognized that, as a consequence of such strength contrast, a highly compacted limestone may not exhibit any signs of breakage in shells, foraminifera, or ooids (Shinn et al., 1977; Bhattacharyya and Friedman, 1979; Shinn and Robbin, 1983). Accordingly the apparent lack of fossil breakage should not be construed as evidence that the limestone was not subjected to mechanical compaction, the quantification of which requires a more thorough scrutiny of related damage microstructure in carbonate rock.

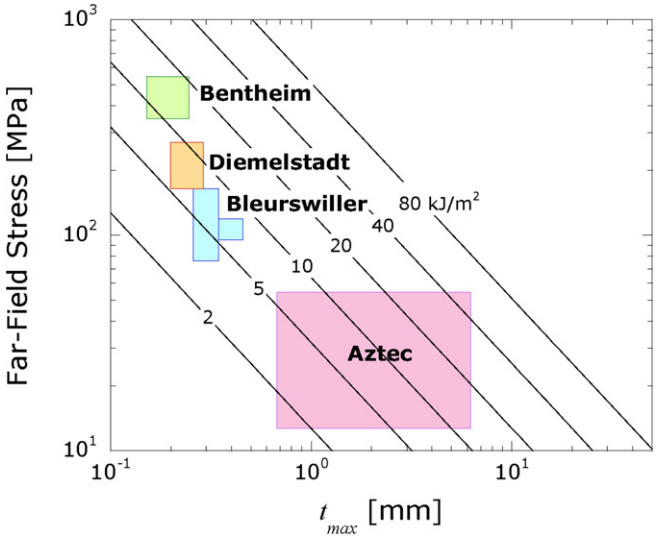


Fig. 22. Predictions of Tembe et al. (2008) for the stress conditions for the propagation of discrete compaction bands in porous sandstones. An extension of Rudnicki's (2007) anti-crack/anti-dislocation model predicts that the far-field stress would scale with the band half-thickness t_{max}. The solid black lines are the stresses predicted by Tembe et al. (2008) assuming a plastic strain of 0.14 for a range of critical strain energy release rates. The colored boxes are laboratory and field data.

There is a paucity of such systematic studies in carbonate formations. In his comprehensive study of compaction in Mississippian skeletal limestones, Meyers (1980) analyzed 390 samples and concluded after close examination that about 90% of the skeletal grainstone and 99% of the skeletal packstones exhibit some compaction features, including grain breakage, plastic grain deformation and grain reorientation. He also demonstrated that the bulk of mechanical compaction in skeletal grains occurs in multicrystalline skeletal grains (bryozoans, brachiopods, ostracodes, solitary corals), whereas deformation of single crystal echinoderms (which are often found in Indiana limestone) is rare. Vajdova et al. (2012) concluded that their observations of twinning, cement damage and allochem rotation in compacted samples of Indiana limestone are qualitatively similar to many of the field examples presented by Meyers (1980).

Field studies of tectonic deformation in carbonate formations have also documented complex interplay of chemical and mechanical compaction processes. In Majella limestone Tondi et al. (2006) concluded that strain localization in deformation bands was associated with pressure solution complemented by cataclasis, whereas Rath et al. (2011) observed primarily cataclastic mechanisms in deformation bands in Leitha limestone. Nevertheless, there is an apparent discrepancy between the field and laboratory observations in relation to the development of compaction bands. Unlike siliciclastic rocks, there has not been to date any laboratory evidence of compaction bands in porous limestone. However, Tondi et al. (2006) reported in the Madonna della Mazza quarry a number of bed-parallel deformation bands that lack any evidence of shearing, which they categorized accordingly as compaction bands. A plausible explanation of this apparent discrepancy between laboratory and field observations is that localized compaction may develop more readily in a porous medium that is spatially heterogeneous. In the quarry the more porous samples are often located in relatively thin coarse-grain beds, and it is possible that the compaction bands found in such beds were due to the preferential development of pore collapse at a relatively low critical pressure related to their high porosities. Additionally Tondi et al. (2006) emphasized the importance of pressure solution processes. To synthesize these field and laboratory observations and to resolve some of the apparent discrepancies remain a major challenge in structure geology and experimental rock deformation.

It should be noted that here we have focused primarily on the mechanical issues, and in particular rate-insensitive deformation. In relation to the brittle-ductile transition in porous rocks, there are situations in which the coupling with fluid flow and chemical reaction, as well as creep deformation may influence the inelastic deformation and failure mode. Though beyond the scope of this review, these are topics that deserve to be covered separately in future dedicated reviews.

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References

Andrews, D.J., 2007. Compaction can limit peak vertical velocity at Yucca Mountain. Eos Trans. AGU 88 (52). Fall Meet. Suppl., Abstract S14C-07.

Andrews, D.J., Hanks, T.C., Whitney, J.W., 2007. Physical limits on ground motion at Yucca Mountain. Bull. Seism. Soc. Am. 97, 1771–1792.

Anselmetti, F.S., Luthi, S., Eberli, G.P., 1998. Quantitative characterization of carbonate pore systems by digital image analysis. AAPG Bull. 82, 1815–1836.

Antonellini, M., Aydin, A., 1994. Effect of faulting on fluid flow in porous sandstones: petrophysical properties. Am. Assoc. Petrol. Geol. Bull. 78, 355–377.

Antonellini, M.A., Pollard, D.D., 1995. Distinct element modeling of deformation bands in sandstone. J. Struct. Geol. 17, 1165–1182.

Ashby, M.F., Sammis, C.G., 1990. The damage mechanics of brittle solids in compression. Pure Appl. Geophys. 133, 489–521.

Atkinson, B.K., Meredith, P.G., 1987. Experimental fracture mechanics data for rocks and minerals. In: Atkinson, B.K. (Ed.), Fracture Mechanics of Rock. Academic Press, London, pp. 477–525.

Aydin, A., Johnson, A.M., 1978. Development of faults as zones of deformation bands and as slip surfaces in sandstone. Pure Appl. Geophys. 116, 931–942.

Aydin, A., Borja, R.I., Eichhubl, P., 2006. Geological and mathematical framework for failure modes in granular rock. J. Struct. Geol. 28, 83–98.

Azeemuddin, M., Scott, T.E., Zaman, M., 1994. Variations in Acoustic Wave Velocity Induced during Deformation of Ekofisk Chalk, RMC-94-17. The University of Oklahoma, Norman, OK, p. 27.

Baechle, G.T., Colpaert, A., Eberli, G.P., Weger, R.J., 2008. Effects of microporosity on sonic velocity in carbonate rocks. The Leading Edge (August), 1012–1018.

Baud, P., Schubnel, A., Wong, T.-F., 2000a. Dilatancy, compaction and failure mode in Solnhofen limestone. J. Geophys. Res. 195, 19289–19303.

Baud, P., Zhu, W., Wong, T.-F., 2000b. Failure mode and weakening effect of water on sandstone. J. Geophys. Res. 105, 16371–16390.

Baud, P., Klein, E., Wong, T.-F., 2004. Compaction localization in porous sandstones: spatial evolution of damage and acoustic emission activity. J. Struct. Geol. 26, 603–624.

Baud, P., Vajdova, V., Wong, T.-F., 2006. Shear-enhanced compaction and strain localization: mechanical data and constitutive parameters for porous sandstones. J. Geophys. Res. 111, B12401. http://dx.doi.org/10.1029/2005JB004101.

Baud, P., Vinciguerra, S., David, C., Cavallo, A., Walker, E., Reuschle, T., 2009. Compaction and failure in high porosity carbonates: mechanical data and microstructural observations. Pure Appl. Geophys. 166, 869–898.

Baud, P., Meredith, P.G., Townend, E., 2012. Permeability evolution during triaxial compaction of an anisotropic porous sandstone. J. Geophys. Res. 117, B5. http://dx.doi.org/10.1029/2012JB009176.

Bemer, E., Vincké, O., Longuemare, P., 2004. Geomechanical log deduced from porosity and mineralogical content. Oil, Gas Sci. Tech. - Rev. IFP 59, 405–426.

Bésuelle, P., Rudnicki, J.W., 2004. Localization: shear bands and compaction bands. In: Guéguen, Y., Boutéca, M. (Eds.), Mechanics of Fluid-Saturated Rocks. Elsevier Academic Press, Amsterdam, pp. 219–321.

Bésuelle, P., Desrues, J., Raynaud, S., 2000. Experimental characterization of the localization phenomenon inside a Vosges sandstone in a triaxial cell. Int. J. Rock Mech. Min. Sci. 37, 1223–1237.

Bésuelle, P., Baud, P., Wong, T.-F., 2003. Failure mode and spatial distribution of damage in Rothbach sandstone in the brittle-ductile transition. Pure Appl. Geophys. 160, 851–868.

Bhatt, J.J., Carroll, M.M., Schatz, J.F., 1975. A spherical model calculation for volumetric response of porous rocks. J. Appl. Mech. 42, 363–368.

Bhattacharyya, A., Friedman, G.M., 1979. Experimental compaction of ooids and lime mud and its implication for lithification during burial. J. Sed. Petrol. 49, 1279–1286.

Bjorlykke, K., Hoeg, K., Faleide, J.I., Jahren, J., 2005. When do faults in sedimentary basins leak? Stress and deformation in sedimentary basins; examples from the North Sea and Haltenbanken, offshore Norway. AAPG Bull. 89, 1019–1031.

Blenkinsop, T.G., 2000. Deformation Microstructures and Mechanisms in Minerals and Rocks. Kluwer, Dordrecht.

Borja, R.I., Aydin, A., 2004. Computational modeling of deformation bands in granular media. I. Geological and mathematical framework. Comput. Method. Appl. Mech. Eng. 193, 2667–2698.

Boutéca, M., Sarda, J.-P., Schneider, F., 1996. Subsidence induced by the production of fluids. Rev. Inst. Franc. Petr. 51, 349–379.

Brace, W.F., 1978. Volume changes during fracture and frictional sliding: a review. PAGEOGH 116, 603–614.

Brzesowsky, R., Spiers, C.J., Peach, C.J., Hangx, S.J.T., 2011. Failure behavior of single sand grains: theory versus experiment. J. Geophys. Res. 116, B06205. http://dx.doi.org/10.1029/2010JB008120.

Byerlee, J.D., 1968. Brittle-ductile transition in rocks. J. Geophys. Res. 73, 4741–4750.

Carroll, M.M., 1980. Compaction of dry or fluid-filled porous materials. Jour. Eng. Mech. Proc. ASCE 106 (EM5), 969–990.

Carroll, M.M., 1991. A critical state plasticity theory for porous reservoir rock. In: Massoudi, M., Rajagopal, K.R. (Eds.), Recent Advances in Mechanics of Structured Continua, vol. 117. ASME AMD, pp. 1–5. Book No. G00617.

Casey, M., Butler, R.W.H., 2004. Modelling approaches to understanding fold development: implications for hydrocarbon reservoirs. Mar. Petrol. Geol. 21, 933–946.

Charalampidou, E.-M., Hall, S.A., Stanchits, S., Lewis, H., Viggiani, G., 2011. Characterization of shear and compaction bands in a porous sandstone deformed under triaxial compression. Tectonophysics 503, 8–17.

Chen, W.F., 1984. Soil mechanics, plasticity and landslides. In: Dvorak, G.J., Shield, R.T. (Eds.), Mechanics of Inelastic Materials. Elsevier, Amsterdam, pp. 31–58.

Choquette, P.W., Pray, L.C., 1970. Geologic nomenclature and classification of porosity in sedimentary carbonates. AAPG Bull. 54, 207–250.

Coelho, L., Soares, A.C., Ebecken, N.F.F., Drummond Alves, J.L., Landau, L., 2005. The impact of constitutive modeling of porous rocks on 2-D wellbore stability analysis. J. Petrol. Sci. Eng. 46, 81–100.

Cundall, P.A., Strack, O.D.L., 1979. A discrete numerical model for granular assemblies. Geotechnique 29, 47–65.

Curran, J.H., Carroll, M.M., 1979. Shear stress enhancement of void compaction. J. Geophys. Res. 84, 1105–1112.

Cuss, R.J., Rutter, E.H., Holloway, R.F., 2003. The application of critical state soil mechanics to the mechanical behaviour of porous sandstones. Int. J. Rock Mech. Min. Sci. 40, 847–862.

Das, A., Nguyen, G.D., Einav, I., 2011. Compaction bands due to grain crushing in porous rocks: a theoretical approach based on breakage mechanics. J. Geophys. Res. 116, B08203. http://dx.doi.org/10.1029/2011JB008265.

Dautriat, J., Gland, N., Dimanov, A., Raphanel, J., 2011. Hydromechanical behavior of heterogeneous carbonate rock under proportional triaxial loadings. J.Geophys. Res. 116, B01205. http://dx.doi.org/10.1029/2009JB000830.

David, C., Wong, T.-F., Zhu, W., Zhang, J., 1994. Laboratory measurement of compaction-induced permeability change in porous rock: implications for the generation and maintenance of pore pressure excess in the crust. Pure Appl. Geophys. 143, 425–456.

Davis, R.O., Selvadurai, A.P.S., 2002. Plasticity and Geomechanics. Cambridge University Press, Cambridge, UK.

Desai, C.S., Siriwardane, H.J., 1984. Constitutive Laws for Engineering Materials. Prentice-Hall, Englewood Cliffs, NJ.

DiMaggio, F.L., Sandler, I.S., 1971. Material model for granular soils. J. Eng. Mech. Div., ASCE 97, 935–950.

Dresen, G., Guéguen, Y., 2004. Damage and rock physical properties. In: Guéguen, Y., Boutéca, M. (Eds.), Mechanics of Fluid-Saturated Rocks. Elsevier, Amsterdam, pp. 169–217.

Drucker, D.C., 1951. A more fundamental approach to plastic stress strain relations. In: Proc. First U. S. National Congress Appl. Mechanics, pp. 487–491.

Edmond, J.M., Paterson, M.S., 1972. Volume change during the deformation of rocks at high pressure. Int. J. Rock Mech. Min. Sci. 9, 161–182.

Eichhubl, P., Hooker, J.N., Laubach, S.E., 2010. Pure and shear-enhanced compaction bands in Aztec Sandstone. J. Struct. Geol. 32, 1873–1886.

Einav, I., 2007a. Breakage mechanics – Part I: theory. J. Mech. Phys. Solids 55, 1274–1297.

Einav, I., 2007b. Soil mechanics: breaking ground. Phil. Trans. R. Soc. A 365, 2985–3002.

Elliott, G.M., Brown, T., 1985. Yield of a soft, high porosity rock. Geotech. 35 (4), 413–423.

Evans, J.P., Bradbury, K.K., 2004. Faulting and fracturing of nonwelded Bishop tuff, Eastern California: deformation mechanisms in very porous materials in the Vadose zone. Vadose Zone J. 3, 602–623.

Evans, B., Wong, T.-F., 1985. Shear localization in rocks induced by tectonic deformation. In: Bazant, Z. (Ed.), Mechanics of Geomaterials. John Wiley, NY.

Evans, B., Fredrich, J., Wong, T.-F., 1990. The brittle–ductile transition in rocks: recent experimental and theoretical progress. In: Duba, A.G., Durham, W.B., Handin, J.W., Wang, H.F. (Eds.), The Heard Volume. Am. Geophys. Union, Geophysical Monograph, vol. 56, pp. 1–20.

Fabre, D., Gustkiewicz, J., 1997. Poroelastic properties of limestones and sandstones under hydrostatic conditions. Int. J. Rock Mech. Min. Sci. 34, 127–134.

Fisher, Q.J., Casey, M., Clennell, M.B., Knipe, R.J., 1999. Mechanical compaction of deeply buried sandstones of the North Sea. Mar. Petrol. Geol. 16, 605–618.

Fortin, J., Schubnel, A., Guéguen, Y., 2005. Elastic wave velocities and permeability evolution during compaction of Bleurswiller sandstone. Int. J. Rock Mechan. Min Sci. 42, 873–889.

Fortin, J., Stanchits, S., Dresen, G., Guéguen, Y., 2006. Acoustic emission and velocities associated with the formation of compaction bands in sandstone. J. Geophys. Res. 111, B10203. http://dx.doi.org/10.1029/2005JB003854.

Fortin, J., Schubnel, A., Guéguen, Y., 2007. Effect of pore collapse and grain crushing on ultrasonic velocities and Vp/Vs. J. Geophys. Res. 112. doi:10.129/2005JB004005.

Fossen, H., Schultz, R.A., Zhipton, Z.K., Mair, K., 2007. Deformation bands in sandstone: a review. J. Geol. Soc. London 164, 755–769.

Fossen, H., Schultz, R.A., Torabi, A., 2011. Conditions and implications for compaction band formation in the Navajo sandstone, Utah. J. Struct. Geol. 33, 1477–1490.

Fossum, A.F., Fredrich, J.T., 2000. Cap plasticity models and dilatant and compactive pre-failure deformation. In: Proc. 4th North Am. Rock Mech. Symp. A.A. Balkema, Rotterdam, pp. 1169–1176.

Fowles, J., Burley, S., 1994. Textural and permeability characteristics of faulted, high porosity sandstones. Mar. Pet. Geol. 11, 608–623.

Fredrich, J.T., Evans, B., Wong, T.-F., 1989. Micromechanics of the brittle to plastic transition in Carrara marble. J. Geophys. Res. 94, 4129–4145.
Fredrich, J.T., Deitrick, G.L., Arguello, J.G., deRouffignac, E.P., 2000. Geomechanical modeling of reservoir compaction, surface subsidence, and casing damage at the Belridge diatomite field. SPE Res. Eval. Eng. 3, 348–359.
Griggs, D.T., Turner, F.J., Heard, H.C., 1960. Deformation of rocks at 500° to 800° C. In: Griggs, D., Handin, J. (Eds.), Rock Deformation. Memoir. Geological Society of America, vol. 79, pp. 39–104.
Grueschow, E., 2005. Yield Cap Constitutive Models for Predicting Compaction Localization in High Porosity Sandstone. Unpublished Ph.D. thesis, Northwestern University.
Grueschow, E., Rudnicki, J.W., 2005. Elliptic yield cap constitutive modeling for high porosity sandstone. Int. J. Solids Struct. 42, 4574–4587.
Guéguen, Y., Fortin, J., 2005. Mécanique des roches en géologie: des processus microscopiques au comportement macroscopique. In: Colloque "Microstructure et Propriétés des Matériaux". ENPC.
Guéguen, Y., Palciauskas, V., 1994. Introduction to the Physics of Rocks. Princeton University Press, Princeton.
Guéguen, Y., Dormieux, L., Boutéca, M., 2004. Fundamentals of poromechanics. In: Guéguen, Y., Boutéca, M. (Eds.), Mechanics of Fluid-Saturated Rocks. Elsevier Academic Press, Amsterdam, pp. 1–54.
Gupta, V., Bergström, J., 1998. Compressive failure of rocks by shear faulting. J. Geophys. Res. 103 (B10), 23875–23895.
Gurson, A.L., 1977. Continuum theory of ductile rupture by void nucleation and growth: Part I – Yield criteria and flow rules for porous ductile media. J. Eng. Mat. Tech. 99, 2–15.
Handin, J., Hager, R.V., Friedman, M., Feather, J.N., 1963. Experimental deformation of sedimentary rock under confining pressure: pore pressure effects. Bull. Am. Assoc. Petrol. Geol. 47, 717–755.
Hazzard, J.F., Mair, K., 2003. The importance of the third dimension in granular shear. Geophys. Res. Lett. 30, 1708. http://dx.doi.org/10.1029/2003GL017534.
Hazzard, J.F., Young, R.P., Maxwell, S.C., 2000. Micromechanical modeling of cracking and failure in brittle rocks. J. Geophys. Res. 105, 16683–16697.
Heard, H.C., 1960. Transition from brittle fracture to ductile flow in Solnhofen limestone as a function of temperature, confining pressure and interstitial fluid pressure. In: Griggs, D., Handin, J. (Eds.), Rock Deformation. Memoir. Geological Society of America, vol. 79, pp. 193–226.
Hill, R.E., 1989. Analysis of Deformation Bands in the Valley of Fire state Park, Nevada. Unpublished M.S. thesis, University of Nevada, Las Vegas.
Hirth, G., Tullis, J., 1994. The brittle-plastic transition in experimentally deformed quartz aggregates. J. Geophys. Res. 99, 11731–11748.
Homand, S., Shao, J.F., 2000. Mechanical behaviour of a porous chalk and effect of saturating fluid. Mech. Cohes. -Frict. Mater. 5, 583–606.
Horii, H., Nemat-Nasser, S., 1986. Brittle failure in compression: splitting, faulting and brittle-ductile transition. Phil. Trans. Royal Soc. London 319, 337–374.
Isida, M., Nemat-Nasser, S., 1987. On mechanics of crack growth and its effects on the overall response of brittle porous solids. Acta Metall. 35, 2887–2898.
Issen, K.A., 2000. Conditions for Localized Deformation in Compacting Porous Rocks. Unpublished Ph.D. thesis, Northwestern University.
Issen, K.A., Rudnicki, J.W., 2000. Conditions for compaction bands in porous rock. J. Geophys. Res. 105, 21529–21536.
Jaeger, J.C., Cook, N.G.W., Zimmerman, R.W., 2007. Fundamentals of Rock Mechanics, fourth ed. Blackwell, Oxford.
Jamison, W.R., Stearns, D.W., 1982. Tectonic deformation of Wingate Sandstone, Colorado National Monument. Am. Assoc. Pet. Geol. Bull. 66, 2584–2608.
Jamison, W.R., Teufel, L.W., 1979. Pore volume changes associated with failure and frictional sliding of a porous sandstone. Proc. U. S. Rock Mech. Symp. 20, 163–170.
Jizba, D.L., 1991. Mechanical and Acoustical Properties of Sandstones and Shales. Unpublished Ph.D. thesis, Stanford University.
Johnson, K.L., 1985. Contact Mechanics. Cambridge University Press, Cambridge.
Johnson, J.N., Green, S.J., 1976. The mechanical response of porous media subject to static loads. In: Cowin, S.C., Carroll, M.M. (Eds.), The Effects of Voids on Material Deformation. AMD. A.S.M.E, pp. 93–124.
Karman, T. von, 1911. Festigkeitsversuche unter allseitigem. Druck. Z. Verein. Dtsch. Ing 55, 1749–1757.
Karner, K.L., Chester, F.M., Kronenberg, A.K., Chester, J.S., 2003. Subcritical compaction and yielding of granular quartz sand. Tectonophysics 377, 357–381.
Karner, S.L., Chester, J.S., Chester, F.M., Kronenberg, A.K., Hajash, A.J., 2005. Laboratory deformation of granular quartz sand: implications for the burial of clastic rocks. AAPG Bull. 89, 603–625.
Katsman, R., Aharonov, E., Scher, H., 2005. Numerical simulation of compaction bands in high-porosity sedimentary rock. Mech. Mat. 37, 143–162.
Kemeny, J.M., Cook, N.G.W., 1991. Micromechanics of deformation in rocks. In: Shah, S.P. (Ed.), Toughening Mechanisms in Quasi-Brittle Materials. Klewer Academic, pp. 155–188.
Klein, E., Baud, P., Reuschlé, T., Wong, T.-F., 2001. Mechanical behaviour and failure mode of Bentheim sandstone under triaxial compression. Phys. Chem. Earth (A) 26, 21–25.
Knipe, R.J., 1997. Juxtaposition and seal diagrams to help analyze fault seals in hydrocarbon reservoirs. AAPG Bull. 81, 187–195.
Kohlstedt, D.L., Evans, B., Mackwell, S.J., 1995. Strength of the lithosphere: constraints imposed by laboratory experiments. J. Geophys. Res. 100, 17587–17602.
Kwasniewski, M., 1989. Laws of brittle failure and of B-D transition in sandstones. In: Maury, V., Fourmaintraux, D. (Eds.), Rock at Great Depth. A.A. Balkema, Rotterdam, pp. 45–58.
Lawn, B.R., 1998. Indentation of ceramics with spheres: a century after Hertz. J. Am. Ceram. Soc. 81, 1977–1994.
Li, L., Holt, R.M., 2002. Particle scale reservoir mechanics. Oil, Gas Sci. Tech. - Rev. IFP 57, 525–538.
Lockner, D.A., Morrow, C.A., 2008. Energy dissipation in Calico Hills tuff due to pore collapse. Eos Trans. AGU 89 (53). Fall Meet. Suppl., Abstract T51A-1856.
Louis, L., Wong, T.-F., Baud, P., Tembe, S., 2006. Imaging strain localization by X-ray computed tomography: discrete compaction bands in Diemelstadt sandstone. J. Struct. Geol. 28, 762–775.
Louis, L., Baud, P., Wong, T.-F., 2009. Microstructural inhomogeneity and mechanical anisotropy associated with bedding in sandstone. Pure Appl. Geophys. 166, 1063–1087.
Lubarda, V.A., Mastilovic, S., Knap, J., 1996. Brittle-ductile transition in porous rocks by cap model. J. Eng. Mech. 122, 633–642.
Mair, K., Elphick, S.C., Main, I.G., 2002. Influence of confining pressure on the mechanical and structural evolution of laboratory deformation band. Geophys. Res. Lett. 29 (10). http://dx.doi.org/10.1029/2001GL013964
Makowitz, A., Milliken, K.L., 2003. Quantification of brittle deformation in burial compaction, Frio and Mount Simon Formation sandstones. Jour. Sed. Res. 73, 1007–1021.
Marketos, G., Bolton, M.D., 2009. Compaction bands simulated is discrete element models. J. Struct. Geol. 31, 479–490.
Mavko, G., Mukerji, T., Dvorkin, J., 1998. The Rock Physics Handbook: Tools for Seismic Analysis in Porous Media. Cambridge University Press, Cambridge.
Menéndez, B., Zhu, W., Wong, T.-F., 1996. Micromechanics of brittle faulting and cataclastic flow in Berea sandstone. J. Struct. Geol. 18, 1–16.
Meyers, W.J., 1980. Compaction in Mississippian skeletal limestones, southwestern New Mexico. J. Sed. Petrol. 50, 457–474.
Mogi, K., 1966. Pressure dependence of rock strength and transition from brittle fracture to ductile flow. Bull. Earthquake Res. Inst. Tokyo Univ. 44, 215–232.
Mollema, P.N., Antonellini, M.A., 1996. Compaction bands: a structural analog for anti-mode I cracks in Aeolian sandstone. Tectonophysics 267, 209–228.
Mowar, S., Zaman, M., Stearns, D.W., Roegiers, J.-C., 1994. Pore collapse mechanisms in Corboba Cream limestone. Proc. North. Am. Rock Mech. Symp. 1, 767–773.
Muir Wood, D., 1990. Soil Behaviour and Critical State Soil Mechanics. Cambridge University Press, Cambridge.
Muir Wood, D., 2007. The magic of sand – The 20th Bjerrum Lecture. Can. Geotech. J. 44, 1329–1350.
Muir Wood, D., Maeda, K., 2008. Changing grading of soil: effect on critical states. Acta Geotech. 3, 3–14.
Myer, L.R., Kemeny, J., Cook, N.G.W., Ewy, R., Suarez, R., Zheng, Z., 1992. Extensile cracking in porous rock under differential compressive stress. Appl. Mech. Rev. 45, 263–280.
Nagel, N.B., 2001. Compaction and subsidence issues within the petroleum industry: from Wilmington to Ekofisk and beyond. Phys. Chem. Earth (A) 26, 3–14.
Nguyen, G.D., Einav, I., 2009. The energetics of cataclasis based on breakage mechanics. Pure Appl. Geophys. 166, 1693–1724.
Olsson, W.A., 1999. Theoretical and experimental investigation of compaction bands in porous rock. J. Geophys. Res. 104, 7219–7228.
Olsson, W.A., Holcomb, D.J., 2000. Compaction localization in porous rock. Geophys. Res. Lett. 27, 3537–3540.
Olsson, W.A., Peng, S.S., 1976. Microcrack nucleation in marble. Int. J. Rock Mech. Min. Sci. 13, 53–59.
Papamichos, E., Tronvoll, J., Skjærstein, A., Unander, T.E., Vardoulakis, I., Sulem, J., 2000. Constitutive testing of Red Wildmoor sandstone. Mech. Cohes. -Frict. Mater. 5, 1–40.
Passchier, C.E., Trouw, R.A.J., 2005. Microtectonics, second ed. Springer, Berlin.
Paterson, M.S., 1958. Experimental deformation and faulting in Wombeyan marble. Bull. Geol. Soc. Am. 69, 465–476.
Paterson, M.S., Wong, T.-F., 2005. Experimental Rock Deformation - The Brittle Field, second ed. Spinger-Verlag, New York.
Pestman, B.J., van Munster, J.G., 1996. An acoustic emission study of damage development and stress-memory effects in sandstone. Int. J. Rock Mech. Min. Sci. 33, 585–593.
Pittman, E.D., 1971. Microporosity in carbonate rocks. AAPG Bull. 55, 1873–1878.
Poirier, J.P., 1980. Shear localization and shear instability in materials in the ductile field. J. Struct. Geol. 2, 135–142.
Potyondy, D.O., Cundall, P.A., 2004. A bonded-particle model for rock. Int. J. Rock Mech. Min. Sci. 41, 1329–1364.
Potyondy, D.O., Cundall, P.A., Lee, C., 1996. Modeling rock using bonded assemblies of circular particles. Proc. North Am. Rock Mech. Symp. 2, 1937–1944.
Rath, A., Exner, U., Tschegg, C., Grasemann, B., Laner, R., Draganits, E., 2011. Diagenetic control of deformation mechanisms in deformation bands in a carbonate grainstone. AAPG Bull. 95, 1369–1381.
Riley, P.R., Goodwin, L.B., Lewis, C.J., 2010. Controls on fault damage zone width, structure, and symmetry in the Bandelier Tuff, New Mexico. J. Struct. Geol. 32, 766–780.
Risnes, R., Kristensen, C.N., Andersen, M.A., 1996. Triaxial tests on high porosity chalk with different saturating fluids. In: Fifth North Sea Chalk Symposium. Reimes, France, p. 12.
Rowe, K.J., Rutter, E.H., 1990. Paleostress estimation using calcite twinning: experimental calibration and application to nature. J. Struct. Geol. 12, 1–17.
Rudnicki, J.W., 2002. Conditions for compaction and shear bands in a transversely isotropic material. Int. J. Solids Struct. 39, 3741–3756.

Rudnicki, J.W., 2007. Models for compaction band propagation. In: David, C., LeRavalec, M. (Eds.), Rock Physics and Geomechanics in the Study of Reservoirs and Repositories. Geological Society of London Special Publication, vol. 284, pp. 107–126.
Rudnicki, J.W., Rice, J.R., 1975. Conditions for the localization of deformation in pressure sensitive dilatant materials. J. Mech. Phys. Solids 23, 371–394.
Rutter, E.H., 1986. On the nomenclature of mode of failure transitions in rocks. Tectonophysics 122, 381–387.
Rutter, E.H., Hadizadeh, J., 1991. On the influence of porosity on the low-temperature brittle-ductile transition in siliciclastic rocks. J. Struct. Geol. 13, 609–614.
Rutter, E.H., Llana-Funez, S., Brodie, K.H., 2009. Dehydration and deformation of intact cylinders of serpentinite. J. Struct. Geol. 31, 29–43.
Sammis, C.G., Ashby, M.F., 1986. The failure of brittle porous solids under compressive stress states. Acta Metall. 34, 511–526.
Schatz, J.F., 1976. Models of inelastic volume deformation for porous geologic materials. In: Cowin, S.C., Carroll, M.M. (Eds.), The Effects of Voids on Material Deformation. AMD. A.S.M.E, pp. 141–170.
Schock, R.N., Heard, H.C., Stevens, D.R., 1973. Stress-strain behavior of a granodiorite and two graywackes on compression to 20 kilobars. J. Geophys. Res. 78, 5922–5941.
Schofield, A.N., Wroth, C.P., 1968. Critical State Soil Mechanics. McGraw Hill, New York.
Scholz, C.H., 2002. The Mechanics of Earthquakes and Faulting, second ed. Cambridge University Press, Cambridge.
Schöpfer, M.P.J., Abe, S., Childs, C., Walsh, J.J., 2009. The impact of porosity and crack density on the elasticity, strength and friction of cohesive granular materials: insights from DEM modeling. Int. J. Rock Mech. Min. Sci. 46, 250–261.
Schultz, R.A., 2009. Scaling and paleodepth of compaction bands, Nevada and Utah. J. Geophys. Res. 114, B03407. http://dx.doi.org/10.1029/2008JB005876.
Schultz, R.A., Fossen, H., 2008. Terminology for structural discontinuities. AAPG Bull. 92, 853–867.
Schultz, R.A., Siddharthan, R., 2005. A general framework for the occurrence and faulting for deformation bands in porous granular rocks. Tectonophysics 411, 1–18.
Schultz, R.A., Okubo, C.H., Fossen, H., 2010. Porosity and grain size controls on compaction band formation in Jurassic Navajo sandstone. Geophys. Res. Lett. 37, L22306. http://dx.doi.org/10.1029/2010GL044909.
Senseny, P.E., 1986. A microcrack model for the deformation and strength of Algerie granite. Proc. U.S. Rock Mechanics Symp. 27, 11–15.
Sheldon, H.A., Barnicoat, A.C., Ord, A., 2006. Numerical modelling of faulting and fluid flow in porous rocks: an approach based on critical state soil mechanics. J. Struct. Geol. 28, 1468–1482.
Shinn, E.A., Robbin, D.M., 1983. Mechanical and chemical compaction in fine-grained shallow-water limestones. J. Sed. Petrol. 53, 595–618.
Shinn, E.A., Halley, R.B., Hudson, J.H., Lidz, B.H., 1977. Limestone compaction: an enigma. Geology 5, 21–24.
Shipton, Z.K., Cowie, P.A., 2001. Damage zone and slip-surface evolution over μm to km scales in high-porosity Navajo sandstone, Utah. J. Struct. Geol. 23, 1825–1844.
Stefanov, Y.P., Chertov, M.A., Aidagulov, G.R., Myasnikov, A.V., 2011. Dynamics of inelastic deformation of porous rocks and formation of localized compaction zones studied by numerical modeling. J. Mech. Phys. Solids. 59, 2323–2340.
Sternlof, K.R., Rudnicki, J.W., Pollard, D.D., 2005. Anticrack-inclusion model for compaction bands in sandstone. J. Geophys. Res. 110, B11403. http://dx.doi.org/10.1029/2005JB003764.
Tembe, S., Vajdova, V., Wong, T.-F., Zhu, W., 2006. Initiation and propagation of strain localization in circumferentially notched samples of two porous sandstones. J. Geophys. Res. 111 (2), B02409. http://dx.doi.org/10.1029/2005JB003611.
Tembe, S., Vajdova, V., Baud, P., Zhu, W., Wong, T.-F., 2007. A new methodology to delineate the compactive yield cap of two porous sandstones under undrained condition. Mech. Mat. 39, 513–523.
Tembe, S., Baud, P., Wong, T.-F., 2008. Stress conditions for the propagation of discrete compaction bands in porous sandstone. J. Geophys. Res. 113, B09409. http://dx.doi.org/10.1029/2007JB005439.
Timoshenko, S., Goodier, J.N., 1951. Theory of Elasticity. McGraw Hill, New York.
Tondi, E., Antonellini, M., Aydin, A., Marchegiani, L., Cello, G., 2006. The role of deformation bands, stylolites and sheared stylolites in fault development in carbonate grainstones of Majella Mountain, Italy. J. Struct. Geol. 28, 376–391.
Townend, E., Thompson, B.D., Benson, P.M., Meredith, P.G., Baud, P., Young, R.P., 2008. Imaging compaction band propagation in Diemelstadt sandstone using acoustic emission locations. Geophys. Res. Lett. 35, L15301. http://dx.doi.org/10.1029/2008GL034723.
Tullis, J., Yund, R.A., 1992. The brittle-ductile transition in feldspar aggregates: an experimental study. In: Evans, B., Wong, T.-f. (Eds.), Fault Mechanics and Transport Properties of Rocks. Academic, San Diego, Calif, pp. 89–117.
Turner, F.J., Griggs, D.T., Heard, H.C., 1954. Experimental deformation of calcite crystals. Geol. Soc. Am. Bull. 65, 883–954.
Underhill, J.R., Woodcock, N.H., 1987. Faulting mechanisms in high-porosity sandstones; new red sandstone, Arran, Scotland. In: Jones, M.E., Preston, R.M.F. (Eds.), Deformation of Sediments and Sedimentary Rocks. Special Publication. Geol. Soc., London, pp. 91–105.
Vajdova, V., Wong, T.-F., 2003. Incremental propagation of discrete compaction bands: acoustic emission and microstructural observations on circumferentially notched samples of Bentheim sandstone. Geophys. Res. Lett. 30 (14), 1775. http://dx.doi.org/10.1029/2003GL017750.
Vajdova, V., Baud, P., Wong, T.-F., 2004a. Compaction, dilatancy and failure in porous carbonate rocks. J. Geophys. Res. 109, B05204. http://dx.doi.org/10.1029/2003JB002508.
Vajdova, V., Baud, P., Wong, T.-F., 2004b. Permeability evolution during localized deformation in Bentheim sandstone. J. Geophys. Res. 109, B10406. http://dx.doi.org/10.1029/2003JB002942.
Vajdova, V., Zhu, W., Chen, T.-M.N., Wong, T.-F., 2010. Micromechanics of brittle faulting and cataclastic flow in Tavel limestone. J. Struct. Geol. 32, 1158–1169.
Vajdova, V., Baud, P., Wu, L., Wong, T.-F., 2012. Micromechanics of inelastic compaction in two allochemical limestones. J. Struct. Geol. 43, 100–117.
Veeken, C.A.M., Walters, J.V., Kenter, C.J., Davis, D.R., 1989. Use of plasticity models for predicting borehole stability. In: Maury, V., Fourmaintraux, D. (Eds.), Rocks at Great Depth, vol. 2. A.A. Balkema, Rotterham, Netherlands, pp. 835–844.
Wang, B., Chen, Y., Wong, T.-F., 2008. A discrete element model for the development of compaction localization in granular rock. J. Geophys. Res. 113, B03202. http://dx.doi.org/10.1029/2006JB004501.
Wimmer, S.A., Karr, D.G., 1996. Compressive failure of microcracked porous brittle solids. Mech. Mat 22, 265–277.
Wong, T.-F., 1990. Mechanical compaction and the brittle–ductile transition in porous sandstones. In: Knipe, R.J., Rutter, R.H. (Eds.), Deformation Mechanisms, Rheology and Tectonics. Geol. Society Special Publication, London, vol. 54, pp. 111–122.
Wong, T.-F., Zhu, W., 1999. Brittle faulting and permeability evolution: hydromechanical measurement, microstructural observation, and network modeling. In: Haneberg, W.C., Mozley, P.S., Moore, C., Goodwin, L.B. (Eds.), Faults and Subsurface Fluid Flow in the Shallow Crust, Geophysical Monograph, vol. 113. American Geophysical Union, Washington, DC, pp. 83–99.
Wong, T.-F., David, C., Zhu, W., 1997. The transition from brittle faulting to cataclastic flow in porous sandstones: mechanical deformation. J. Geophys. Res. 102, 3009–3025.
Wong, T.-F., Baud, P., Klein, E., 2001. Localized failure modes in a compactant porous rock. Geophys. Res. Lett. 28, 2521–2524.
Wong, T.-F., David, C., Menéndez, B., 2004. Mechanical compaction. In: Guéguen, Y., Boutéca, M. (Eds.), Mechanics of Fluid-Saturated Rocks. Elsevier Academic Press, Amsterdam, pp. 55–114.
Wu, X.Y., Baud, P., Wong, T.-F., 2000. Micromechanics of compressive failure and spatial evolution of anisotropic damage in Darley Dale sandstone. Int. J. Rock Mech. Min. Sci. 37, 143–160.
Wu, C.-Y., Ruddy, O.M., Bentham, A.C., Hancock, B.C., Best, S.M., Elliott, J.A., 2005. Modelling the mechanical behaviour of pharmaceutical powders during compaction. Powder Tech. 152, 107–117.
Xiao, X., Evans, B., 2003. Shear-enhanced compaction during non-linear viscous creep of porous calcite-quartz aggregates. Earth Planet. Sci. Lett. 216, 725–740.
Xie, S.Y., Shao, J.F., Xu, W.Y., 2011. Influences of chemical degradation on mechanical behaviour of a limestone. Int. J. Rock Mech. Min. Sci. 48, 741–747.
Yarushina, V., Podladchikov, Y., 2010. Low-frequency attenuation due to pore-scale inelasticity. Geophysics 75, 51–63.
Zeuch, D.H., Grazier, J.M., Argüello, J.G., Ewsuk, K.G., 2001. Mechanical properties and shear failure surfaces for two alumina powders in triaxial compression. J. Mat. Sci. 36, 2911–2924.
Zhang, J., Wong, T.-F., Davis, D.M., 1990a. Micromechanics of pressure-induced grain crushing in porous rocks. J. Geophys. Res. 95, 341–352.
Zhang, J., Wong, T.-F., Yanagidani, T., Davis, D.M., 1990b. Pressure-induced microcracking and grain crushing in Berea and Boise sandstones: acoustic emission and quantitative microscopy measurements. Mech. Mater. 9, 1–15.
Zhu, W., 2010. Micromechanics of Failure in Porous Carbonate and Volcanic Rocks. Unpublished Ph.D. thesis, Stony Brook University.
Zhu, W., Baud, P., Wong, T.-F., 2010. Micromechanics of cataclastic pore collapse in limestone. J. Geophys. Res. 115, B04405. http://dx.doi.org/10.1029/2009JB006610.
Zhu, W., Baud, P., Vinciguerra, S., Wong, T.-F., 2011. Micromechanics of brittle faulting and cataclastic flow in Alban Hills tuff. J. Geophys. Res. 116, B06209. http://dx.doi.org/10.1029/2010JB008046.
Zimmerman, R.W., 1991. Compressibility of Sandstones. Elsevier, Amsterdam.